Supplement Information

Contents

1 Explorative Data Analysis

1.1 Endpoint Statistics

	count	mean	std	min	25%	50%	75%	max
Endpoint								
HLM	3087.000000	1.317860	0.624665	0.000000	0.675687	1.203876	1.801643	3.372714
MDR1_ER	2642.000000	0.397285	0.687779	-1.162425	-0.162181	0.152006	0.903767	2.725057
RLM	3054.000000	2.253831	0.753537	0.000000	1.683900	2.309848	2.834440	3.969622
Sol	2173.000000	1.257814	0.684925	-1.000000	1.147985	1.542825	1.687351	2.179264
hPPB	1808.000000	0.643974	0.725078	-1.593460	0.158873	0.678518	1.204120	2.000000
rPPB	885.000000	0.719997	0.749950	-1.638272	0.232996	0.801404	1.278525	2.000000

Table 1: Different statistics of the activity values per Endpoint

1.1.1 Ro5 Characteristics

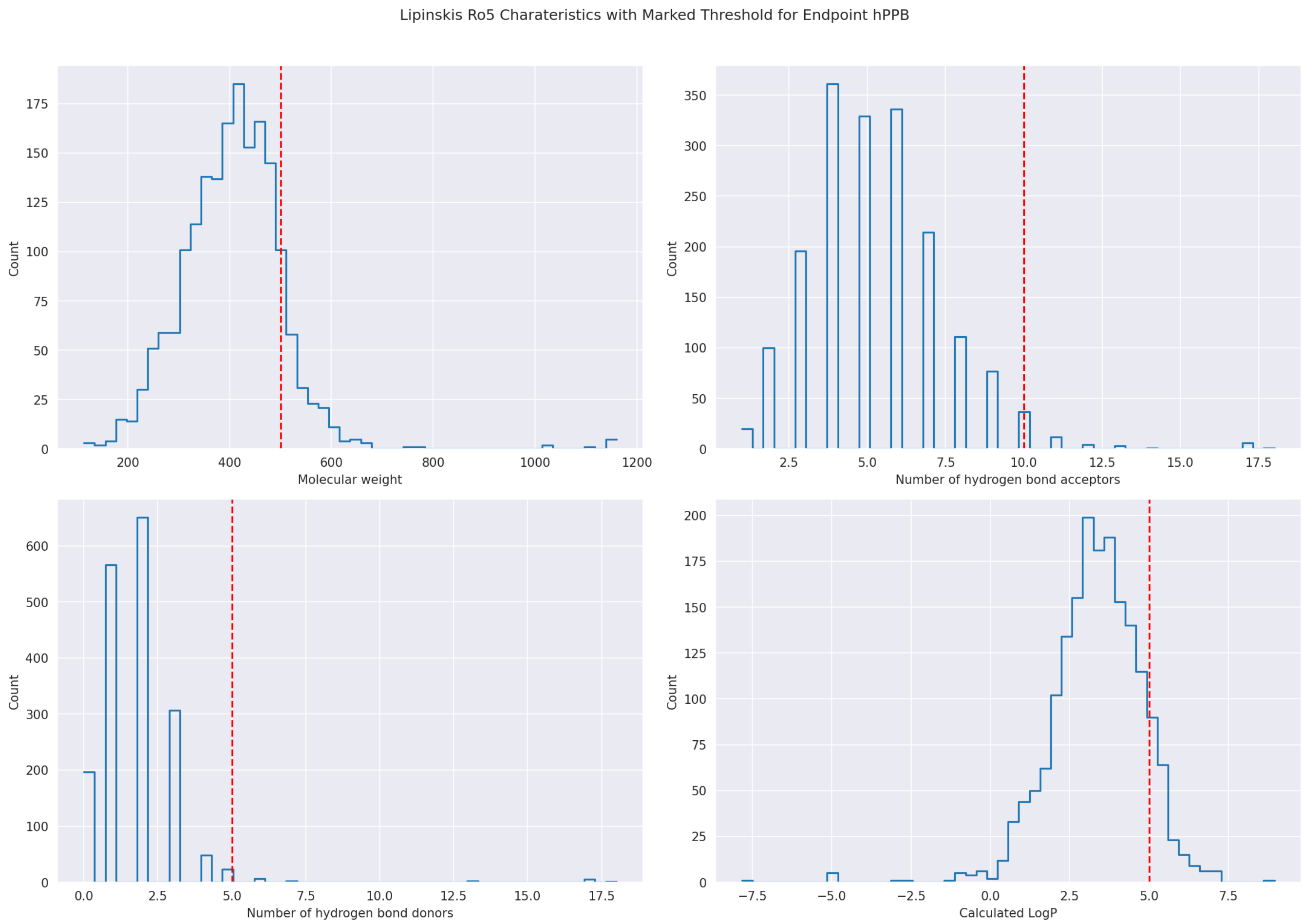


Figure 1: Distribution of ro5 characteristics for endpoint hppb.

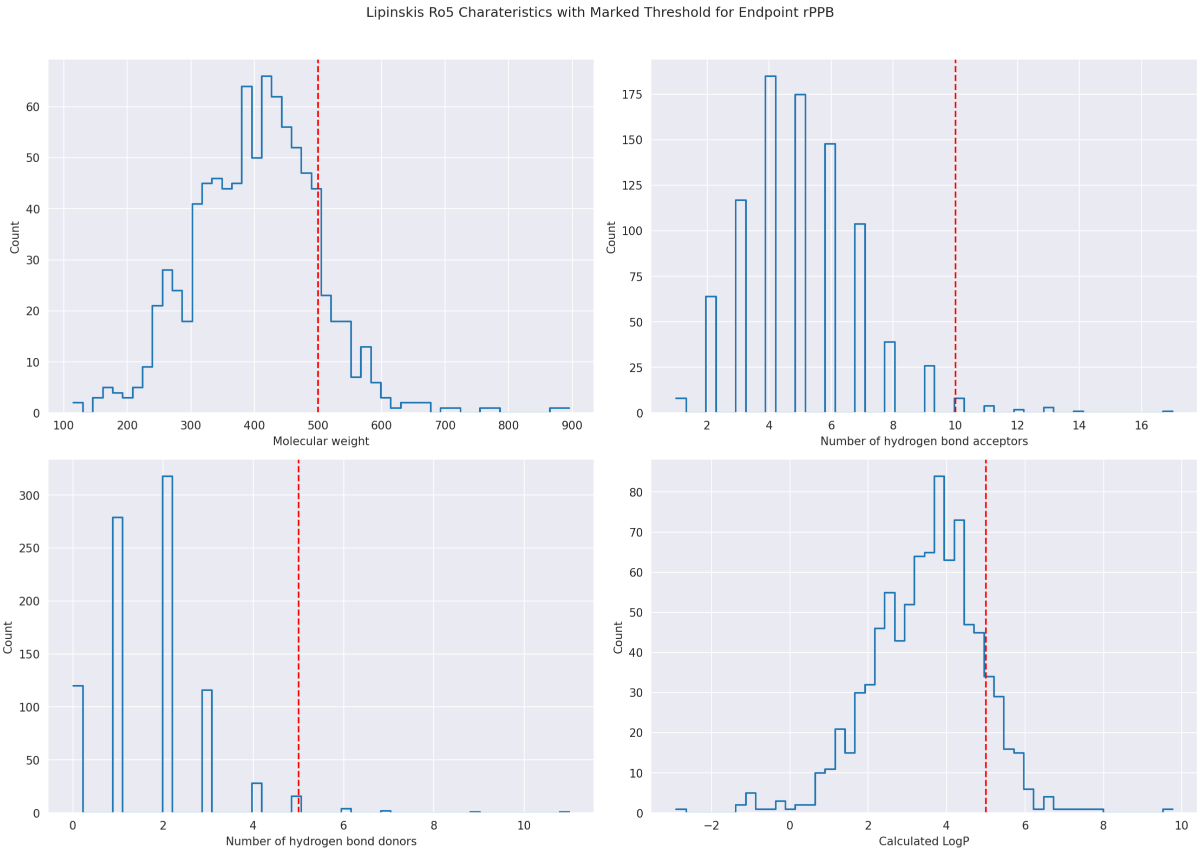


Figure 2: Distribution of ro5 characteristics for endpoint rppb.

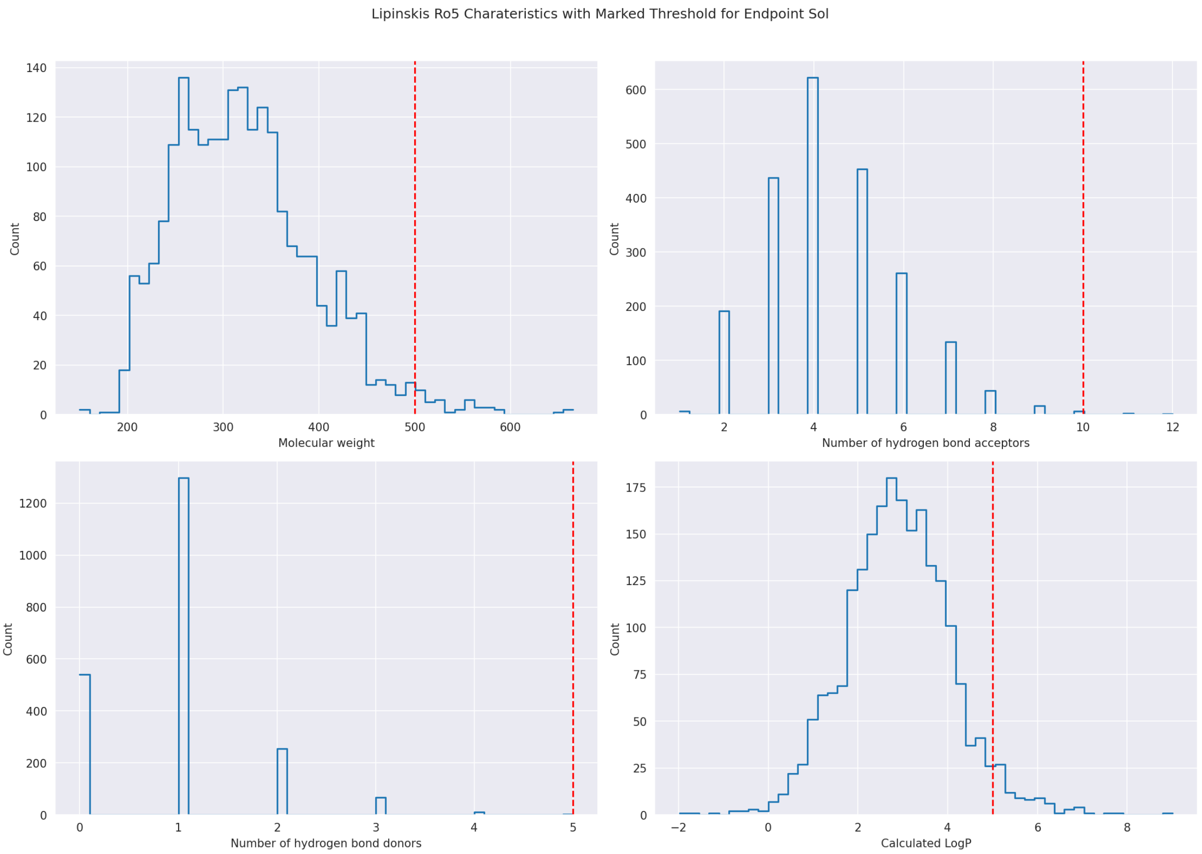


Figure 3: Distribution of ro5 characteristics for endpoint solubility.

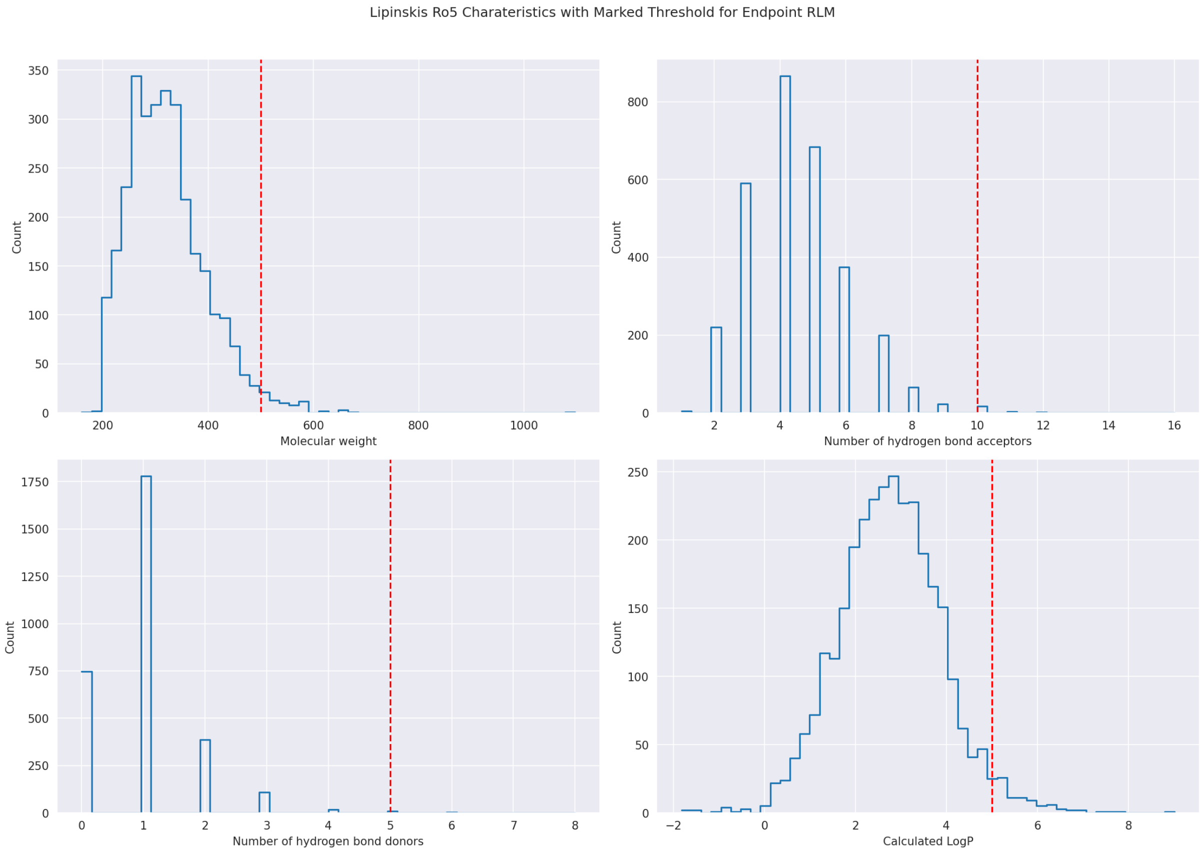


Figure 4: Distribution of ro5 characteristics for endpoint RLM.

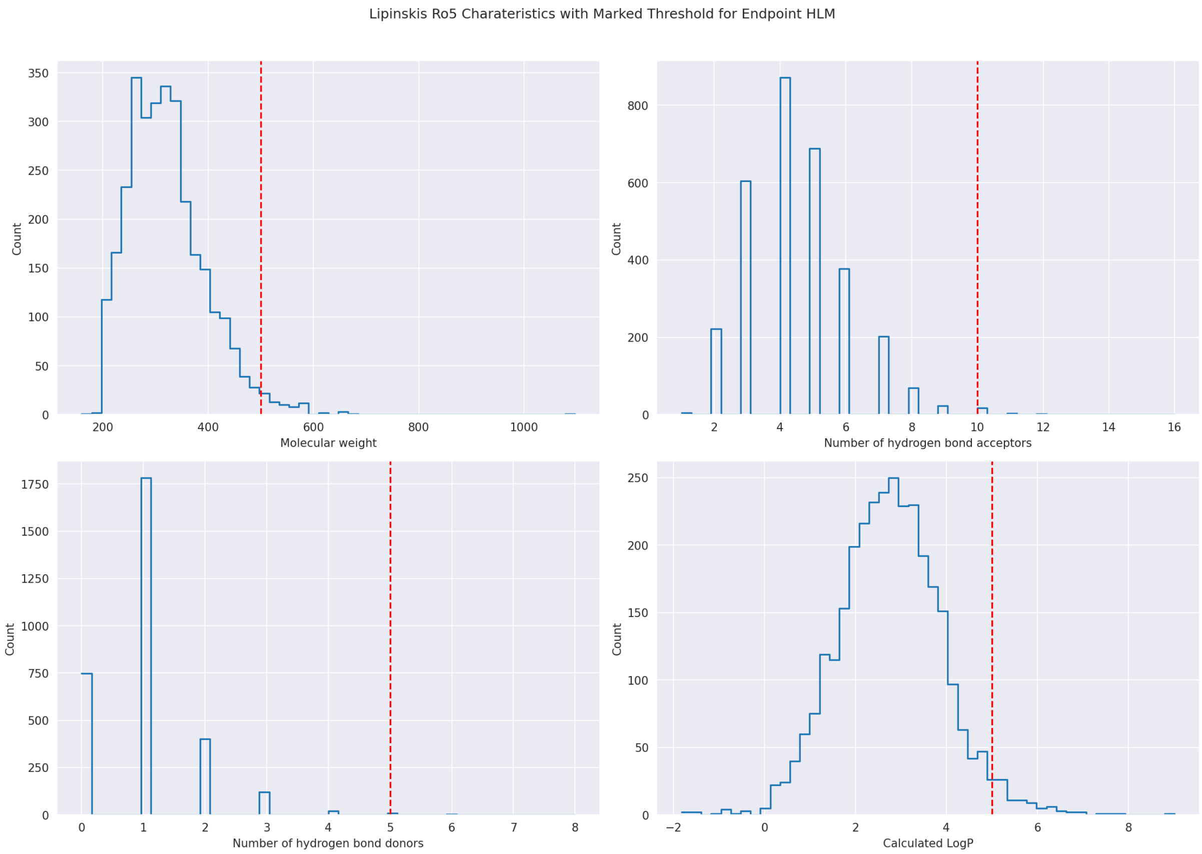


Figure 5: Distribution of ro5 characteristics for endpoint HLM.

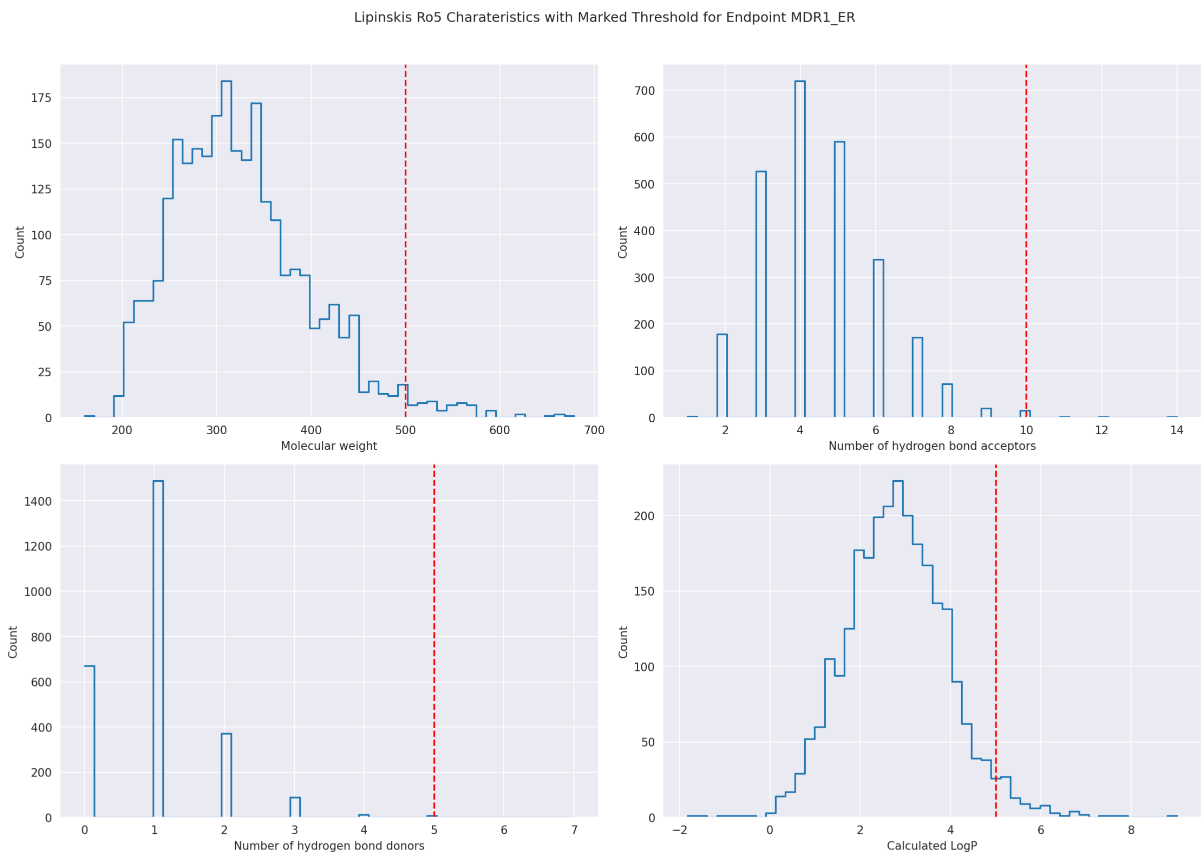
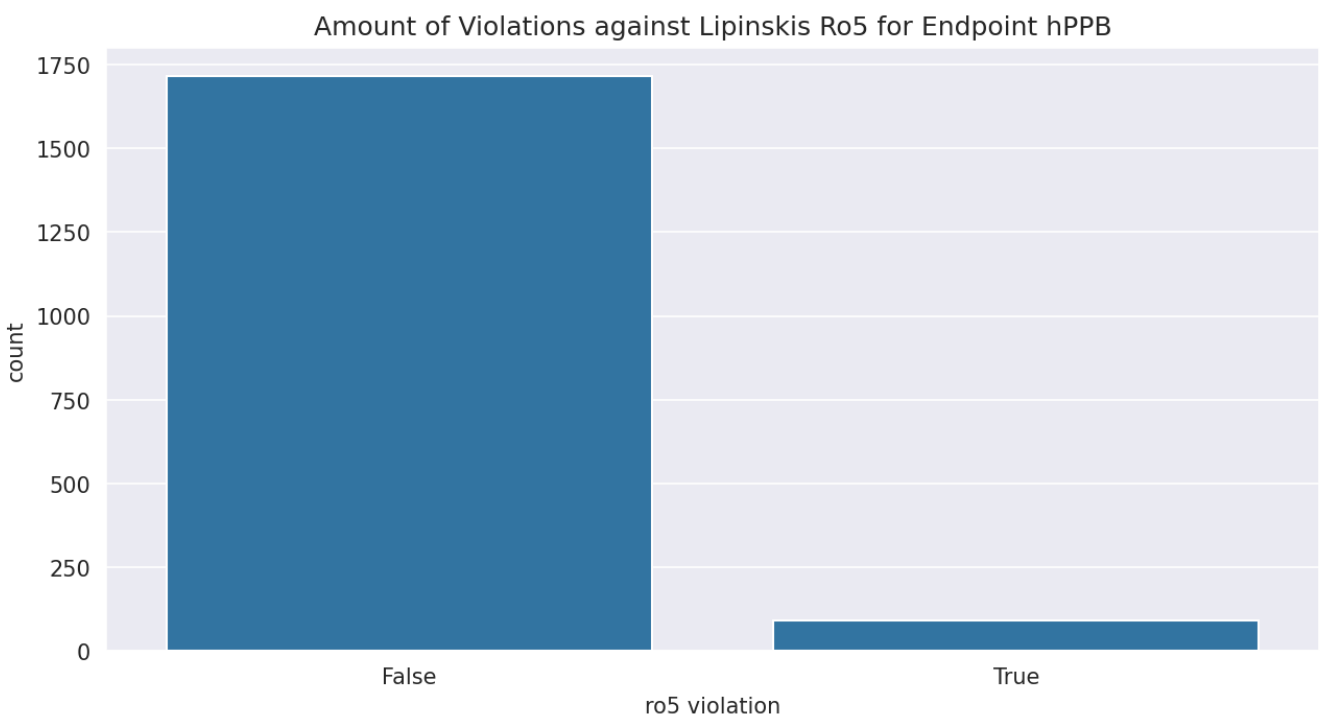
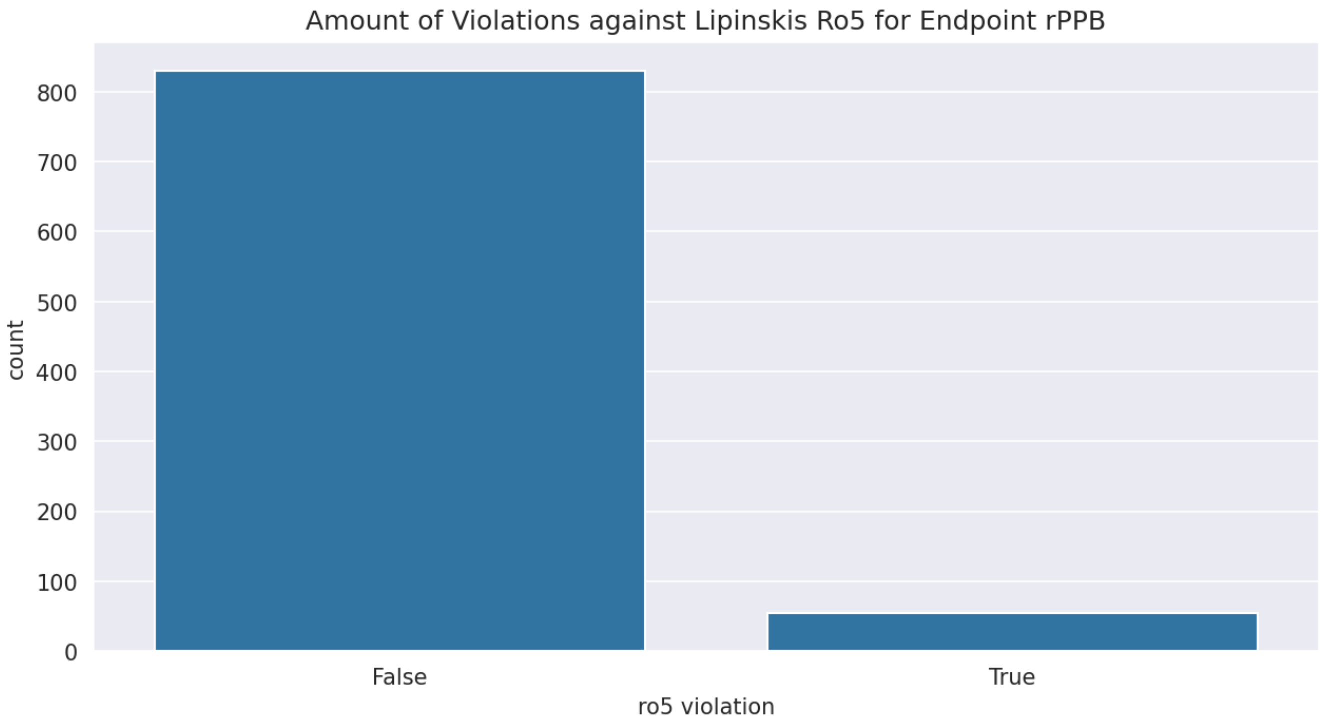
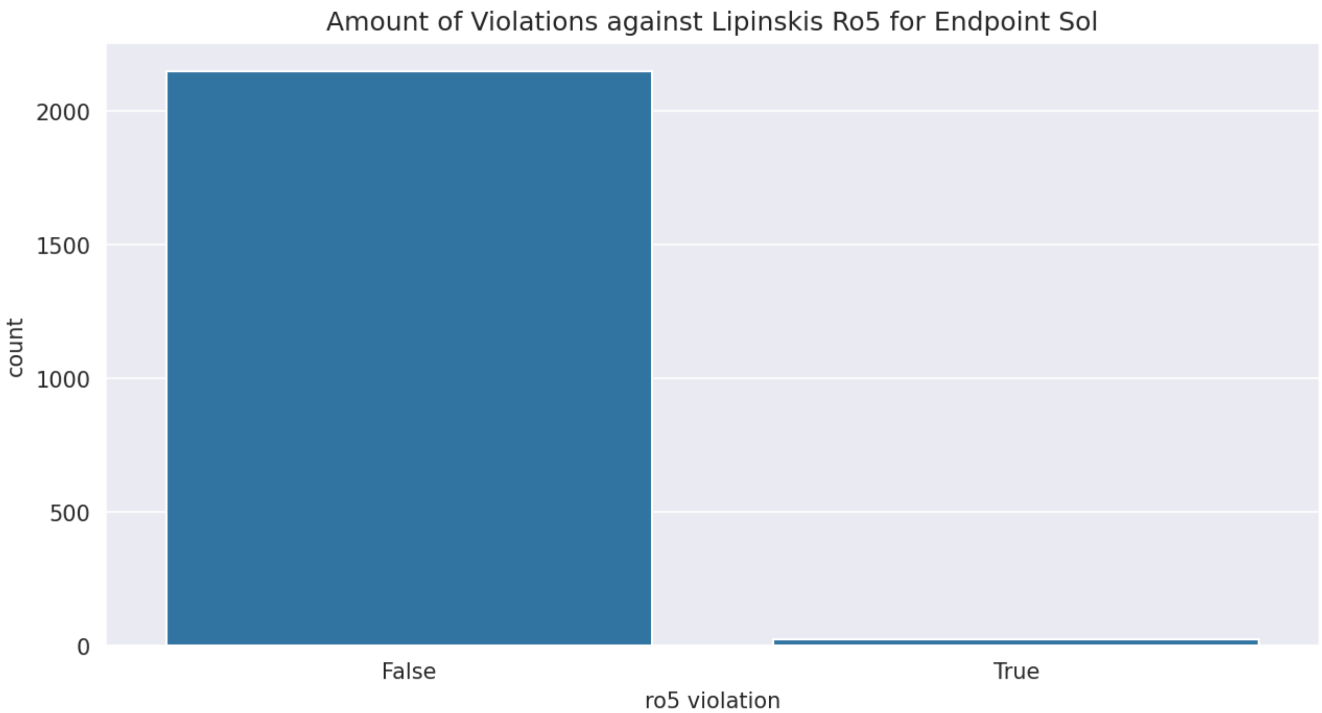


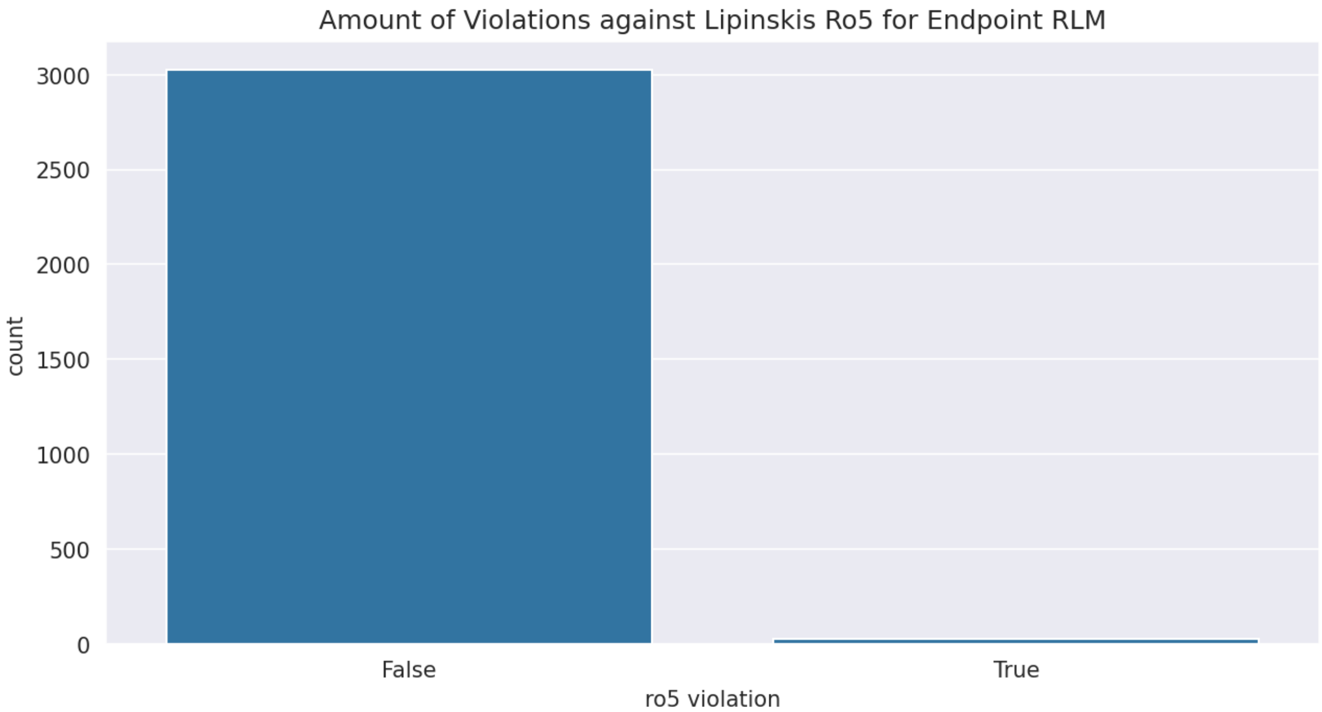
Figure 6: Distribution of ro5 characteristics for endpoint MDR1-ER.

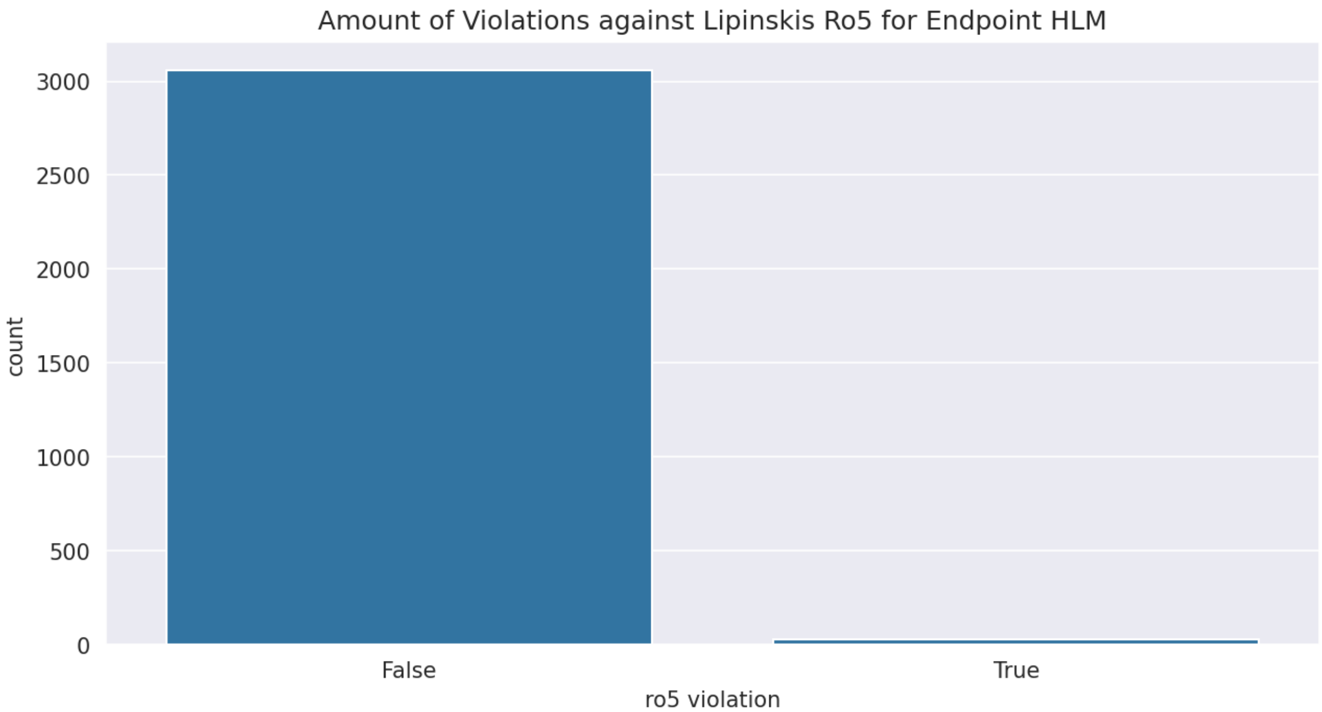
1.1.2 Ro5 Violations

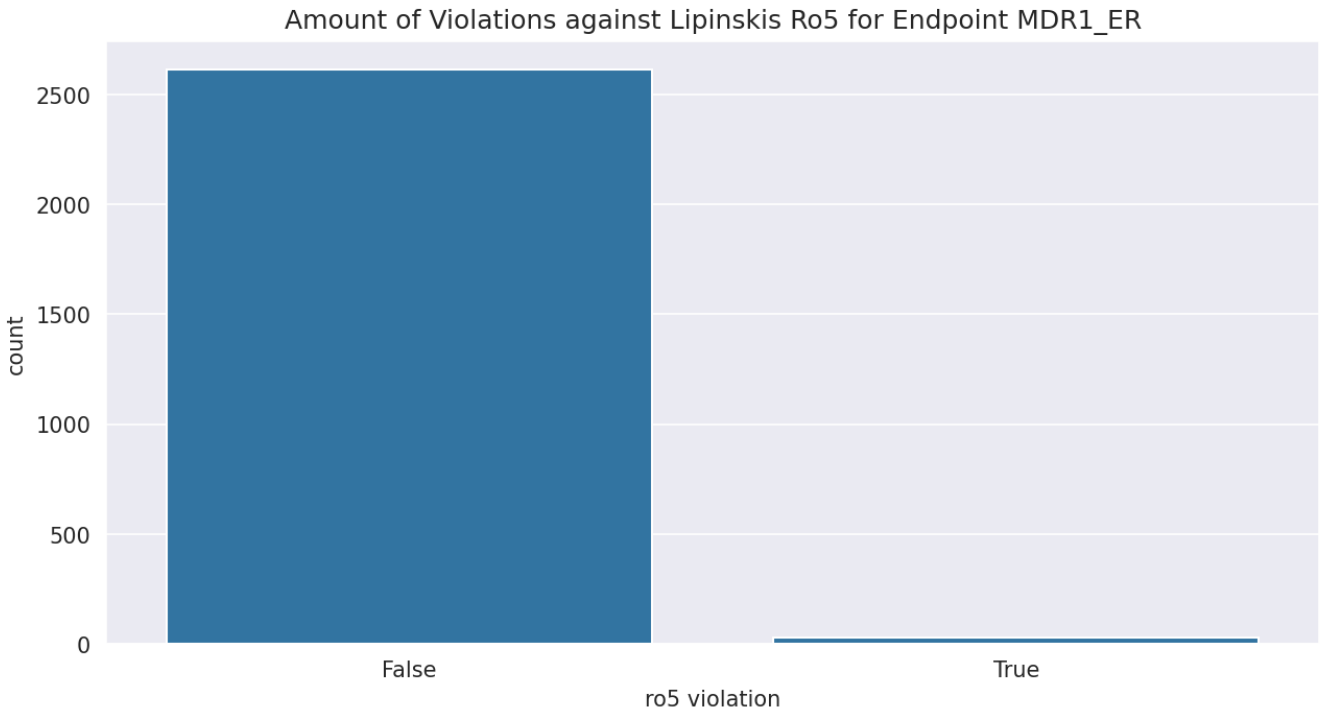



Amount of Violations against Lipinskis Ro5 for Endpoint rPPB
count
800
700
600
500
400
300
200
100
0
False
True
ro5 violation


Amount of Violations against Lipinskis Ro5 for Endpoint Sol
count
2000
1500
1000
500
0
False
True
ro5 violation


Amount of Violations against Lipinskis Ro5 for Endpoint RLM
count
3000
2500
2000
1500
1000
500
0
False
True
ro5 violation

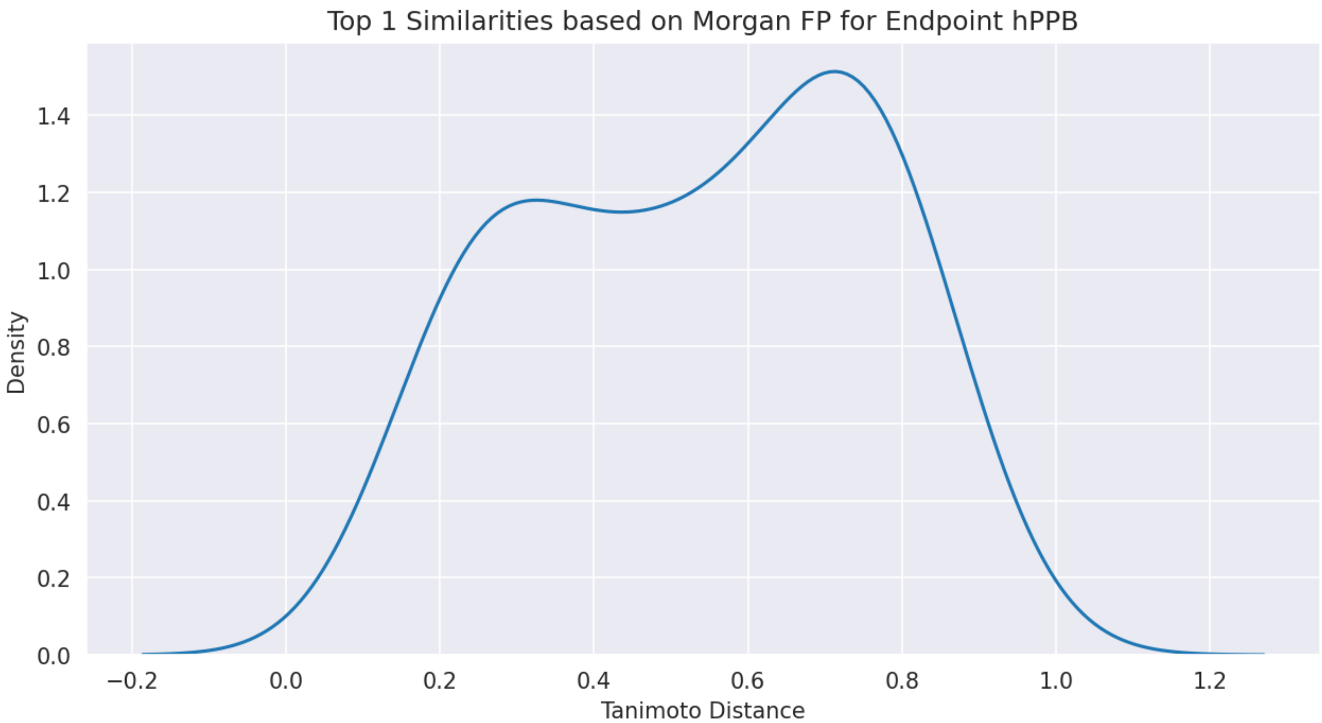

Amount of Violations against Lipinskis Ro5 for Endpoint HLM
count
3000
2500
2000
1500
1000
500
0
False
True
ro5 violation

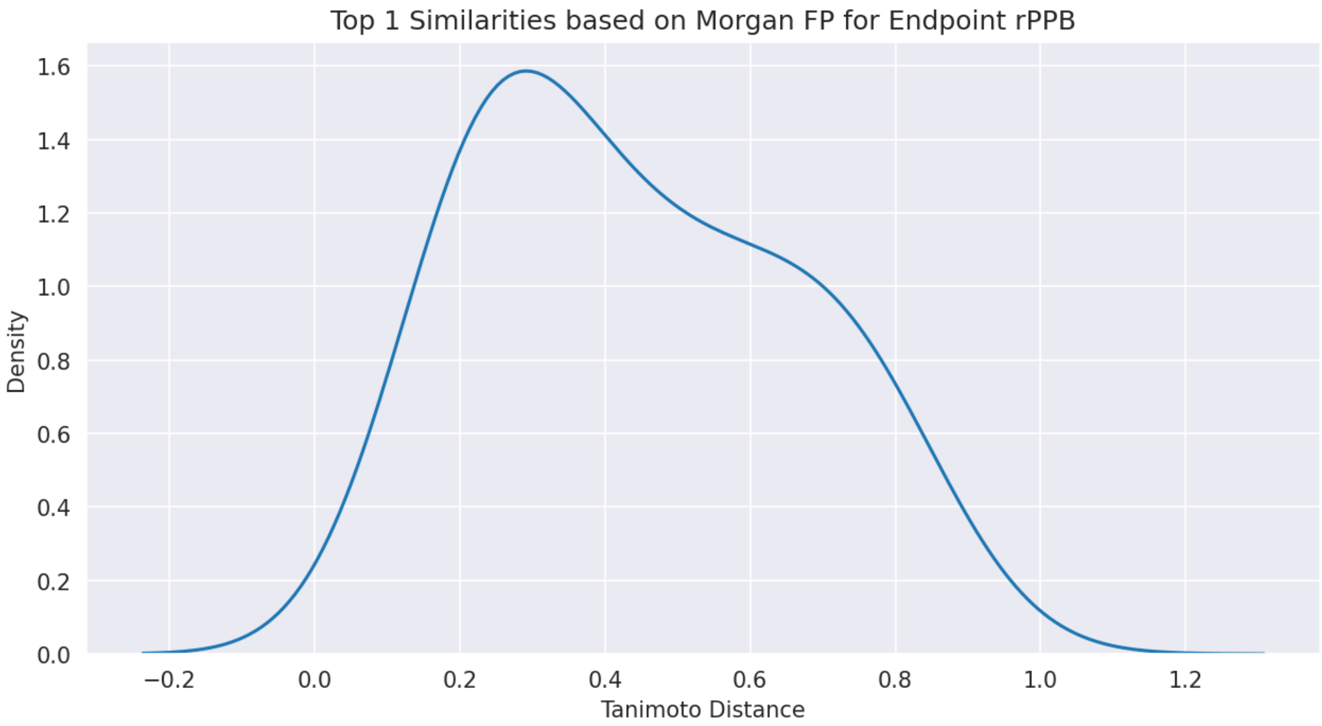


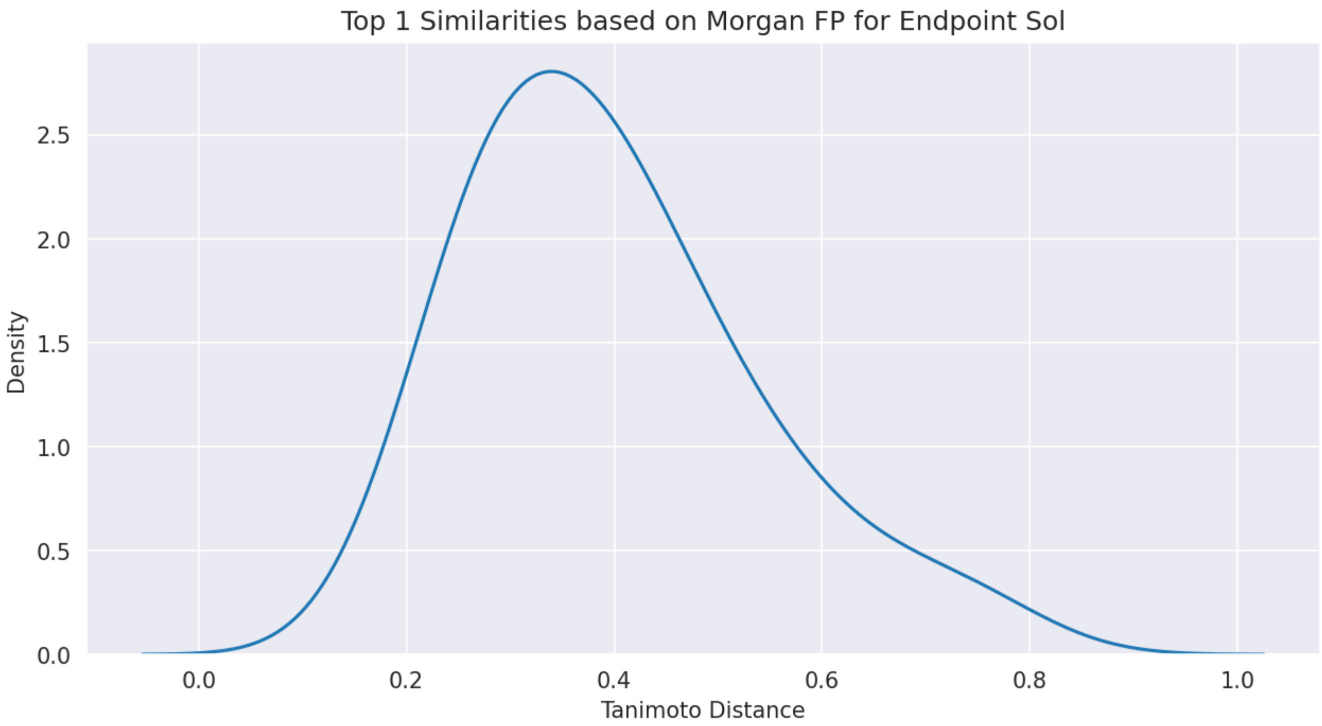
1.2 Similarity Analysis

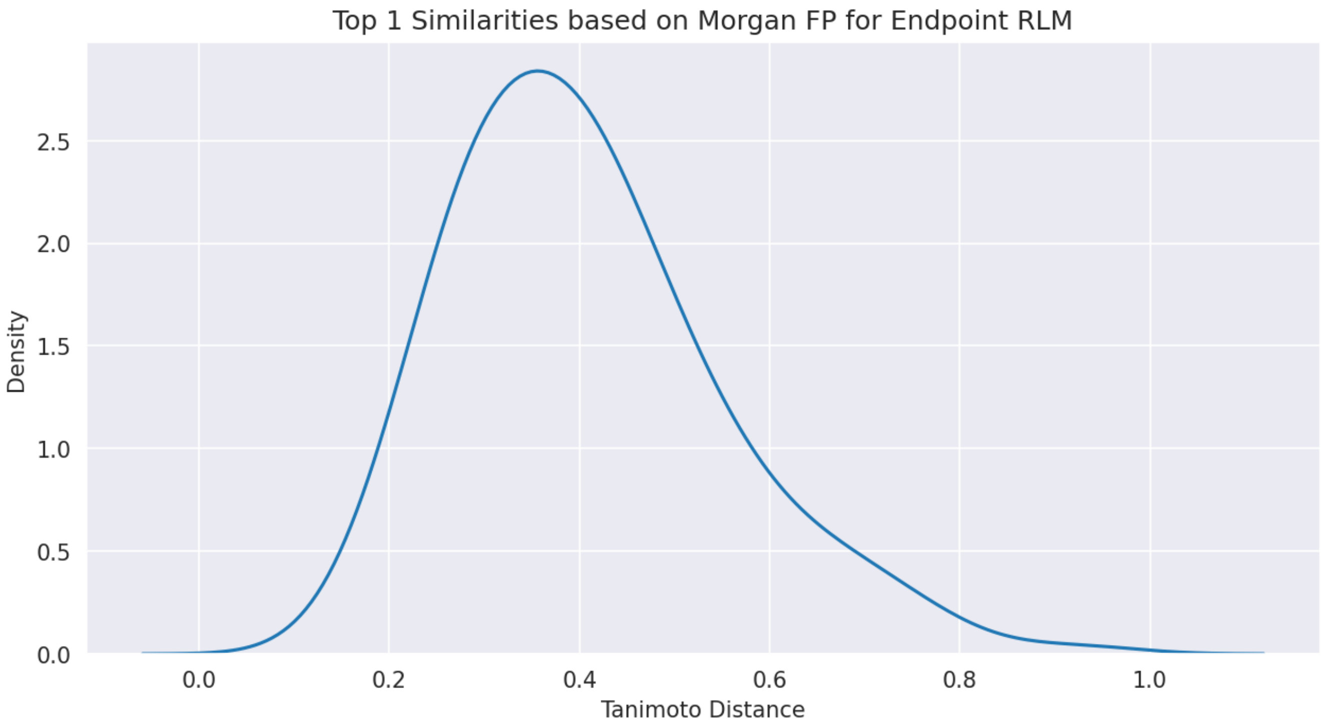
We calculate similarities based on the Tanimoto-distance within the datasets for each endpoint.

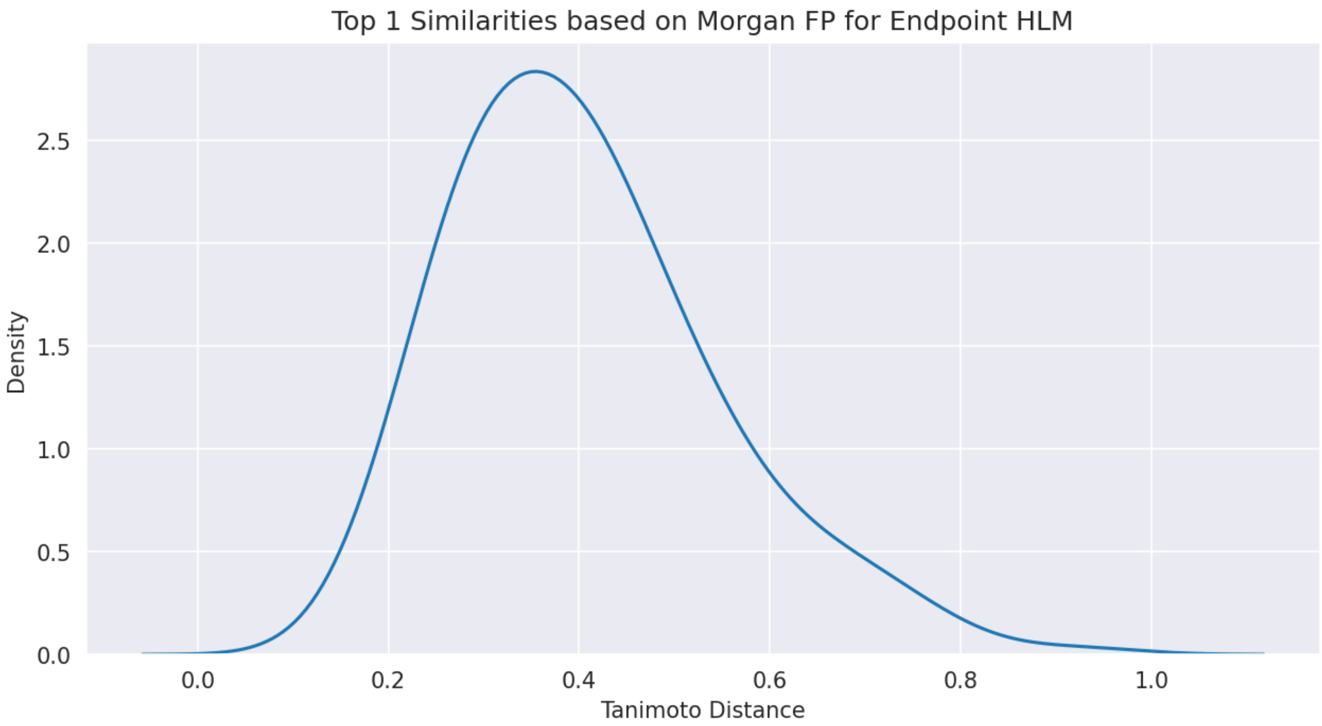
1.2.1 Top 1 Similarities

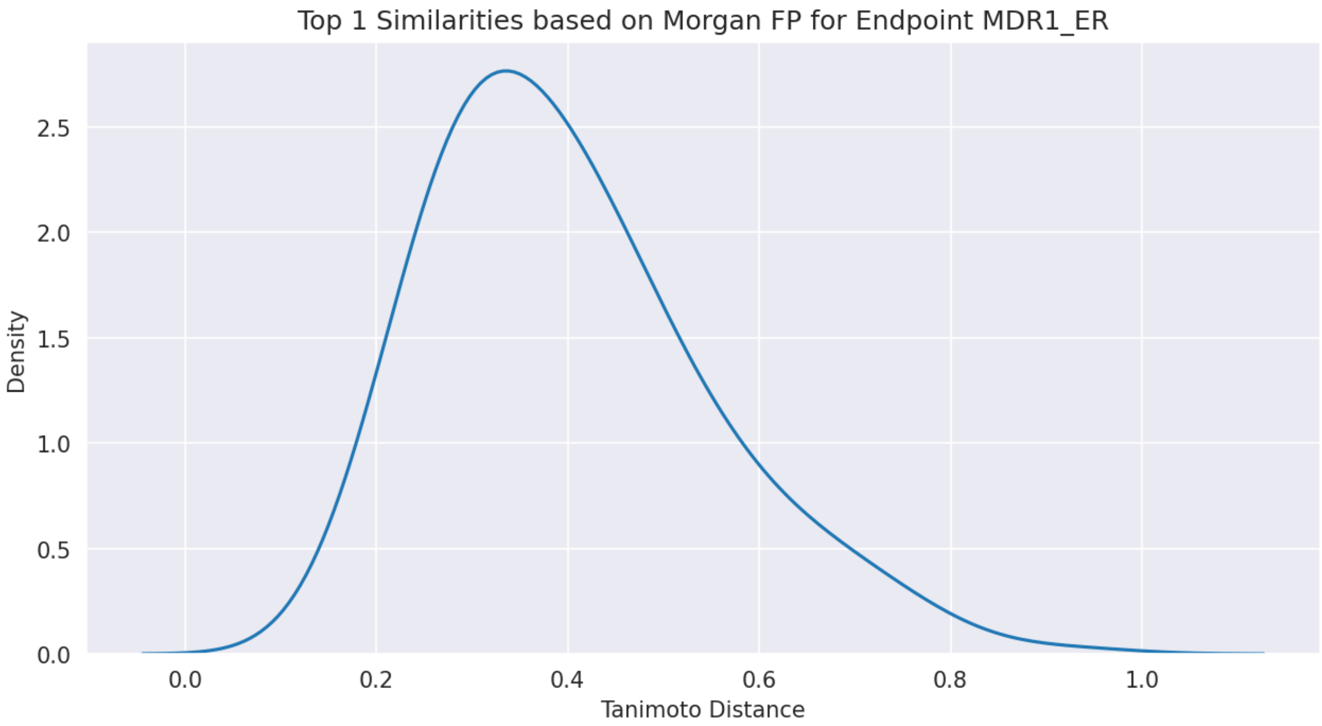



Top 1 Similarities based on Morgan FP for Endpoint rPPB
Density
Tanimoto Distance
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6

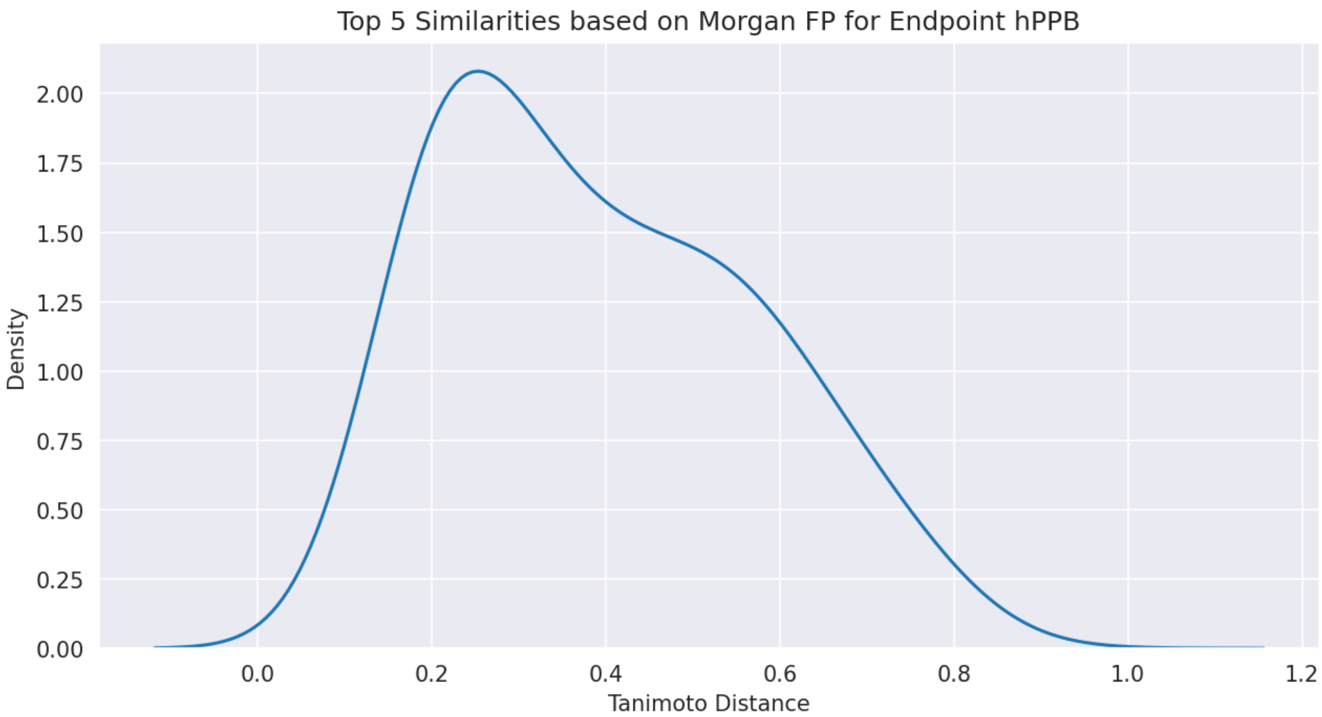

Top 1 Similarities based on Morgan FP for Endpoint Sol
Density
Tanimoto Distance
0.0
0.2
0.4
0.6
0.8
1.0
0.5
1.5
2.0
2.5

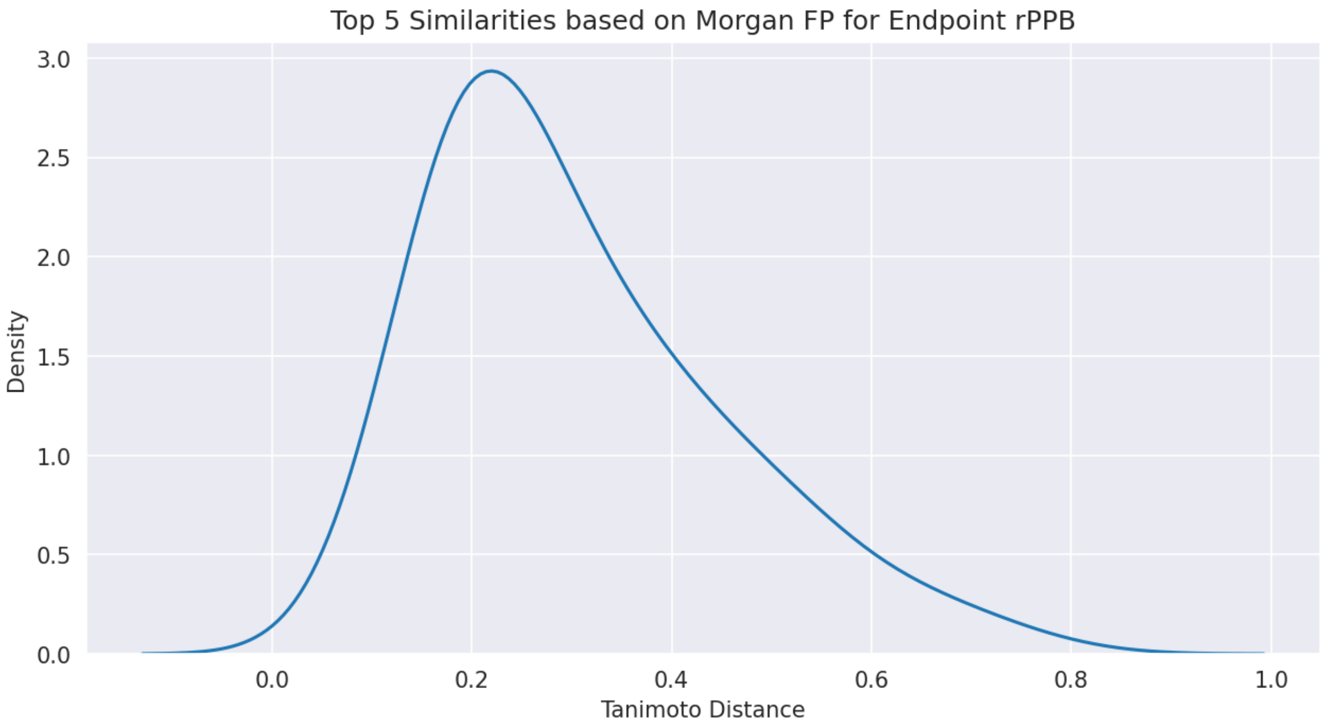

Top 1 Similarities based on Morgan FP for Endpoint RLM
Density
Tanimoto Distance
0.0
0.5
1.0
1.5
2.0
2.5
0.0
0.2
0.4
0.6
0.8
1.0

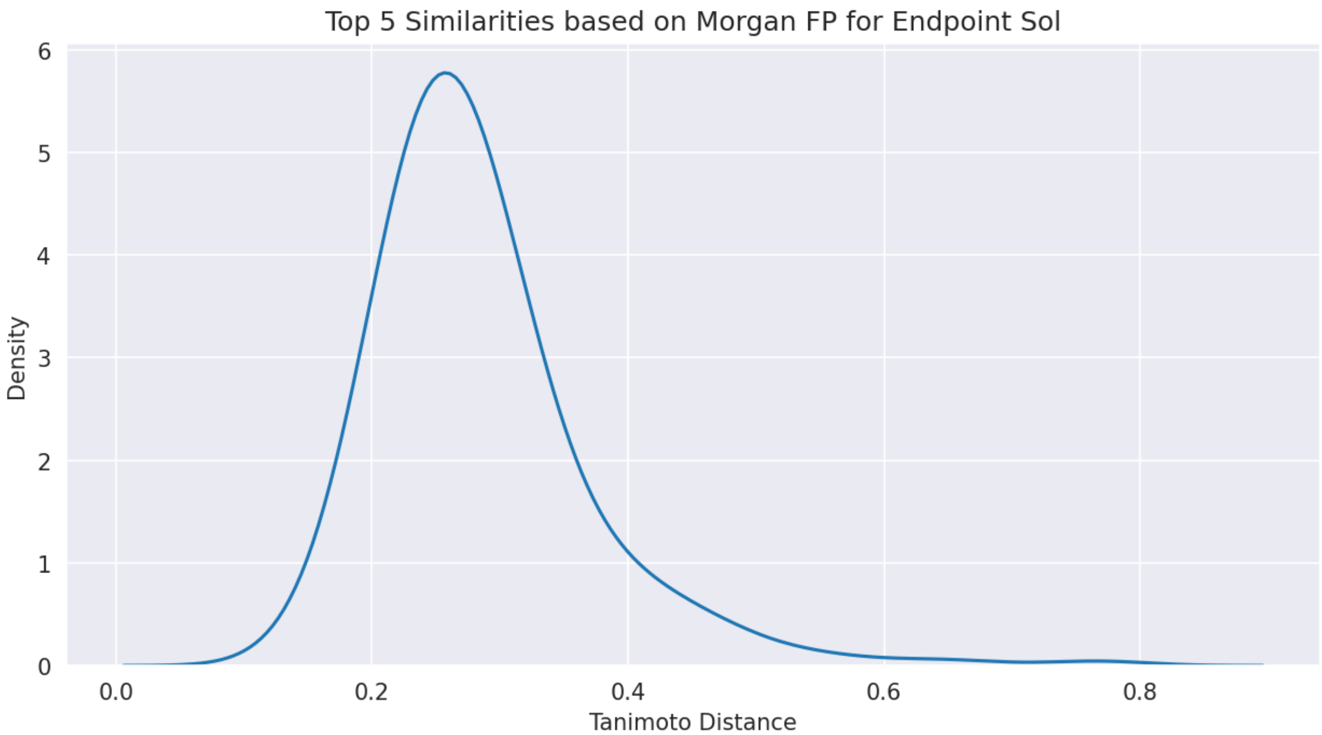

Top 1 Similarities based on Morgan FP for Endpoint HLM
Density
Tanimoto Distance
0.0
0.5
1.0
1.5
2.0
2.5
0.0
0.2
0.4
0.6
0.8
1.0

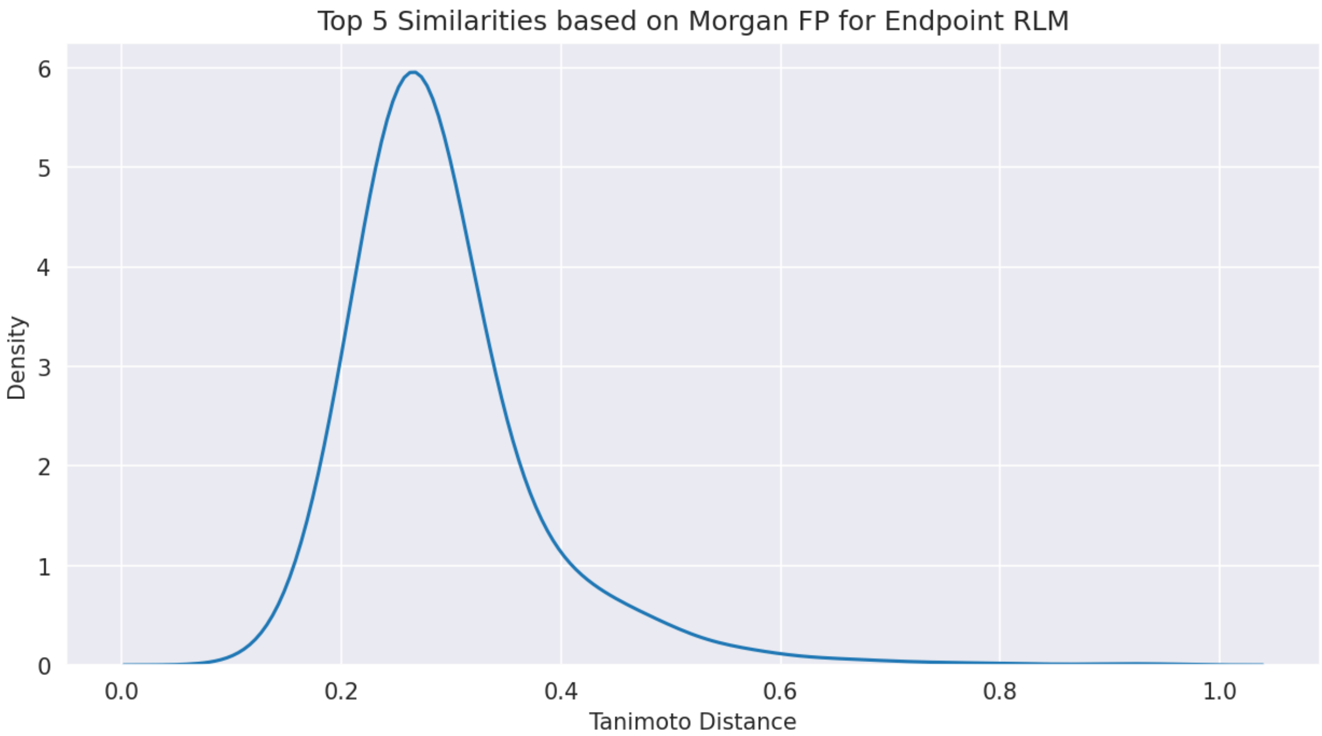


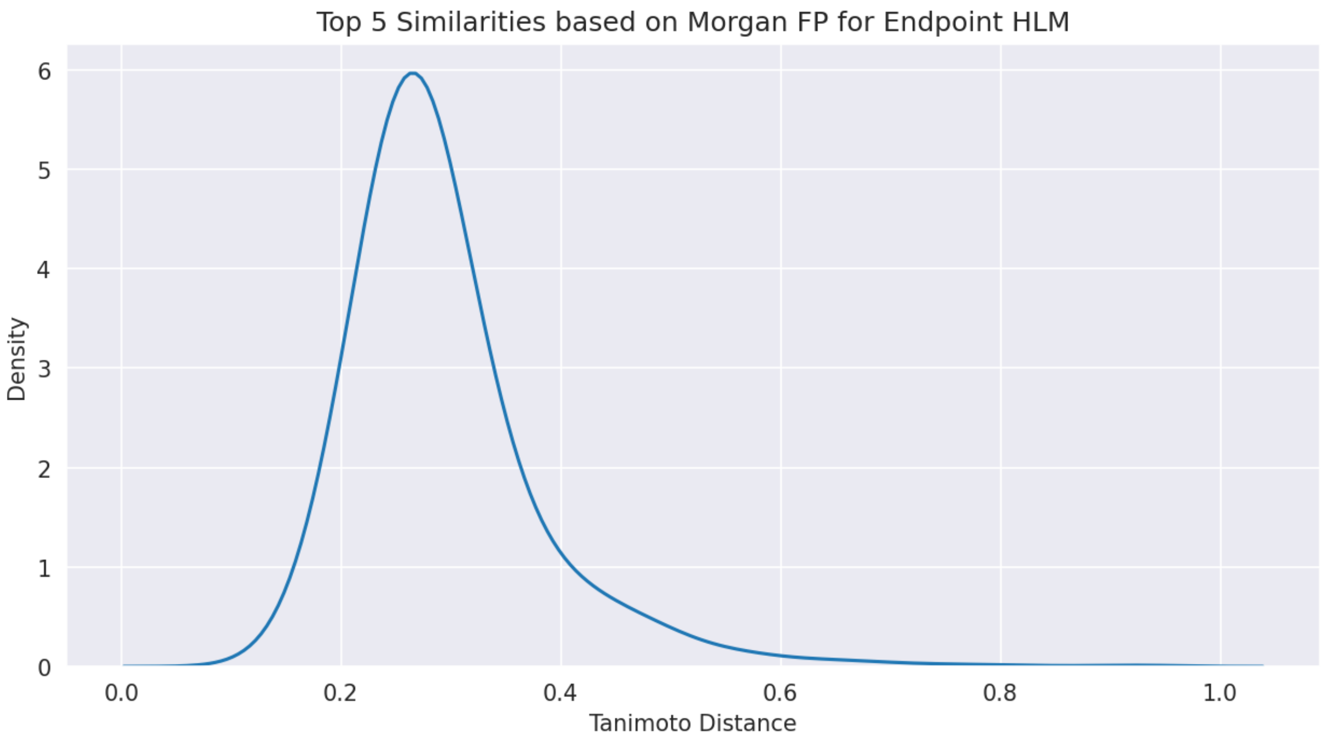
1.2.2 Top 5 Similarities

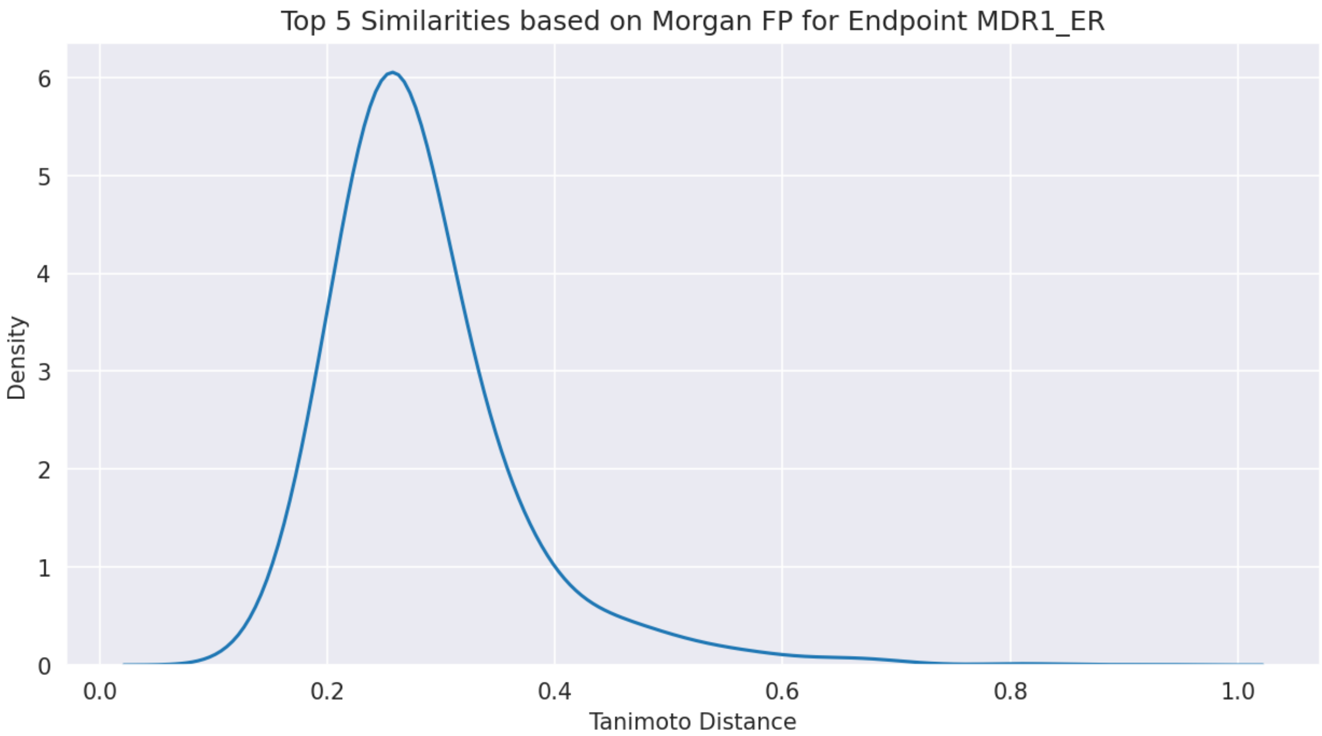



Top 5 Similarities based on Morgan FP for Endpoint rPPB
Density
Tanimoto Distance
0.0
0.2
0.4
0.6
0.8
1.0
0.5
1.0
1.5
2.0
2.5
3.0


Top 5 Similarities based on Morgan FP for Endpoint Sol
Density
Tanimoto Distance
0.0
0.2
0.4
0.6
0.8
0
1
2
3
4
5
6


Top 5 Similarities based on Morgan FP for Endpoint RLM
Density
Tanimoto Distance
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

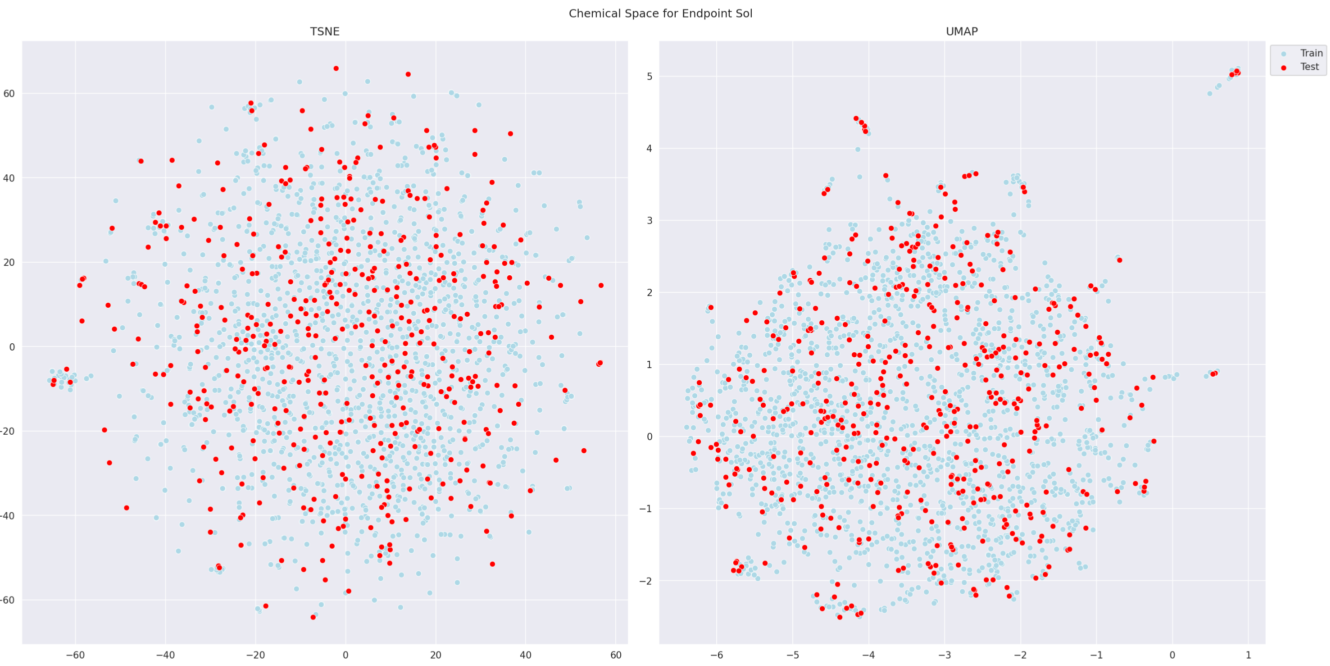

Top 5 Similarities based on Morgan FP for Endpoint HLM
Density
Tanimoto Distance
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0



1.3 Chemical Space

We use Uniform Manifold Approximation and Projection (UMAP) [1] and t-distributed Stochastic Neighbor Embedding (t-SNE) [2] on Morgan-Fingerprints to plot the chemical space. A Principal Component Analysis (PCA) [3] with 90% explained variance to increase the speed of the t-SNE.


Chemical Space for Endpoint rPPB
TSNE
UMAP
Train
Test


Chemical Space for Endpoint Sol
TSNE
UMAP
Train
Test


Chemical Space for Endpoint RLM
TSNE
UMAP
Train
Test


Chemical Space for Endpoint HLM
TSNE
UMAP
Train
Test

2 Hyperparameter Tuning (Optuna)

2.1 Tuned Parameters

2.1.1 MLP

	count	mean	std	min	25%	50%	75%	max
params_batch_size	600.000000	45.680000	37.975534	16.000000	16.000000	32.000000	64.000000	128.000000
params_dropout	600.000000	0.152667	0.151044	0.000000	0.050000	0.100000	0.250000	0.500000
params_lr	600.000000	0.000497	0.000262	0.000100	0.000269	0.000483	0.000707	0.001000
params_n_layers	600.000000	2.286667	0.717877	1.000000	2.000000	2.000000	3.000000	3.000000
params_neurons	600.000000	999.166667	540.187291	100.000000	500.000000	1000.000000	1400.000000	2000.000000

Table 2: Used Hyperparameter with aggregation statistics over all trials and endpoints

2.1.2 XGB

	count	mean	std	min	25%	50%	75%	max
params_colsample_bytree	600.000000	0.764406	0.144059	0.500032	0.653261	0.756095	0.895519	0.999962
params_gamma	600.000000	1.077003	0.813423	0.000762	0.371746	0.921733	1.620666	2.998790
params_learning_rate	600.000000	0.278590	0.272126	0.000177	0.058797	0.172932	0.446394	0.998100
params_max_depth	600.000000	30.453333	20.899341	1.000000	11.000000	29.000000	47.000000	75.000000
params_min_child_weight	600.000000	26.500000	13.438663	1.000000	16.000000	27.000000	37.000000	50.000000
params_n_estimators	600.000000	1201.448333	535.736512	50.000000	796.750000	1279.000000	1672.000000	2000.000000
params_reg_alpha	600.000000	1.147620	0.814739	0.010652	0.439320	1.040181	1.768977	2.997911
params_reg_lambda	600.000000	1.495191	0.821171	0.010037	0.834410	1.563095	2.111239	2.998447
params_subsample	600.000000	0.697816	0.131451	0.500183	0.593060	0.684300	0.785432	0.999770

Table 3: Used Hyperparameter with aggregation statistics over all trials and endpoints

2.1.3 FTT

	count	mean	std	min	25%	50%	75%	max
params_amount_heads	600.000000	3.755000	1.461232	2.000000	2.000000	4.000000	5.000000	7.000000
params_attn_dropout	600.000000	0.246417	0.145780	0.000000	0.150000	0.250000	0.350000	0.500000
params_batch_size	600.000000	45.626667	35.638239	16.000000	16.000000	32.000000	64.000000	128.000000
params_dropout	600.000000	0.248000	0.150445	0.000000	0.150000	0.250000	0.350000	0.500000
params_encoder_output_dim	600.000000	210.855000	54.318488	128.000000	164.000000	207.000000	248.000000	352.000000
params_ff_dropout	600.000000	0.294417	0.134579	0.000000	0.200000	0.300000	0.400000	0.500000
params_head_dim	600.000000	81.680000	29.354905	32.000000	54.000000	85.000000	107.250000	128.000000
params_layer_norm	600.000000	0.400000	0.490307	0.000000	0.000000	0.000000	1.000000	1.000000
params_lr	600.000000	0.001179	0.001931	0.000100	0.000254	0.000447	0.000995	0.009959
params_n_layers	600.000000	1.946667	0.740317	1.000000	1.000000	2.000000	2.000000	3.000000
params_neurons	600.000000	941.500000	568.917565	100.000000	400.000000	900.000000	1400.000000	2000.000000
params_transformer_depth	600.000000	3.876667	1.466870	2.000000	2.000000	4.000000	5.000000	7.000000

Table 4: Used Hyperparameter with aggregation statistics over all trials and endpoints

2.1.4 TabPFN

	count	mean	std	min	25%	50%	75%	max
params_n_estimators	150.000000	52.200000	29.395247	1.000000	28.000000	52.000000	75.000000	100.000000
params_softmax_temperature	150.000000	0.652355	0.255954	0.064033	0.483665	0.720137	0.846330	0.999908

Table 5: Used Hyperparameter with aggregation statistics over all trials and endpoints

2.1.5 NDF

	count	mean	std	min	25%	50%	75%	max
params_batch_size	300.000000	198.400000	63.785594	128.000000	128.000000	256.000000	256.000000	256.000000
params_feature_rate	300.000000	0.598000	0.245282	0.100000	0.400000	0.650000	0.800000	0.950000
params_lr	300.000000	0.000486	0.000242	0.000100	0.000288	0.000464	0.000658	0.001000
params_n_tree	300.000000	59.246667	26.253189	5.000000	39.000000	62.000000	81.000000	100.000000
params_tree_depth	300.000000	2.586667	1.265369	1.000000	2.000000	2.000000	3.250000	5.000000

Table 6: Used Hyperparameter with aggregation statistics over all trials and endpoints

2.2 Best Models per Endpoint

2.2.1 MLP

	0	1	2	3	4	5
params_batch_size	32	16	16	16	16	16
params_dropout	0.000000	0.050000	0.050000	0.000000	0.000000	0.050000
params_lr	0.000998	0.000368	0.000722	0.000752	0.000735	0.000951
params_n_layers	3	3	3	2	3	2
params_neurons	300	1400	1200	700	1700	600
R2	0.550729	0.519987	0.377448	0.547285	0.507441	0.588257
Endpoint	hPPB	rPPB	Sol	RLM	HLM	MDR1_ER

2.2.2 XGB

	0	1	2	3	4	5
params_colsample_bytree	0.666236	0.786335	0.707556	0.705590	0.504046	0.957273
params_gamma	0.061295	0.005656	0.989881	0.197460	0.122448	0.621108
params_learning_rate	0.027672	0.029067	0.011314	0.036516	0.030615	0.032090
params_max_depth	28	39	6	3	48	12
params_min_child_weight	35	48	18	3	32	29
params_n_estimators	968	954	1700	796	1876	1534
params_reg_alpha	0.308862	1.861244	0.414514	0.633075	1.211260	0.323975
params_reg_lambda	2.388754	1.684115	2.018378	1.774816	1.152835	0.299130
params_subsample	0.697376	0.843737	0.501254	0.718638	0.631661	0.668970
R2	0.538259	0.480938	0.324976	0.506176	0.455305	0.523042
Endpoint	hPPB	rPPB	Sol	RLM	HLM	MDR1_ER

2.2.3 FTT

	0	1	2	3	4	5
params_amount_heads	4	5	2	5	2	2
params_attn_dropout	0.350000	0.050000	0.400000	0.200000	0.100000	0.250000
params_batch_size	16	16	32	32	32	32
params_dropout	0.150000	0.350000	0.250000	0.250000	0.500000	0.200000
params_encoder_output_dim	211	152	156	235	153	236
params_ff_dropout	0.200000	0.350000	0.350000	0.300000	0.300000	0.400000
params_head_dim	34	46	109	107	123	88
params_layer_norm	0	0	1	0	0	1
params_lr	0.000713	0.000929	0.000352	0.000392	0.000373	0.000314
params_n_layers	2	1	2	3	1	2
params_neurons	100	1100	1500	400	1100	500
params_transformer_depth	5	6	5	4	2	2
R2	0.498920	0.484245	0.349076	0.488386	0.447629	0.522455
Endpoint	hPPB	rPPB	Sol	RLM	HLM	MDR1_ER

2.2.4 TabPFN

	0	1	2	3	4	5
params_average_before_softmax	False	False	False	False	True	True
params_n_estimators	12	25	37	63	60	71
params_softmax_temperature	0.535125	0.879084	0.985963	0.999908	0.999180	0.990631
R2	0.547659	0.506257	0.355444	0.515052	0.460849	0.559711
Endpoint	hPPB	rPPB	Sol	RLM	HLM	MDR1_ER

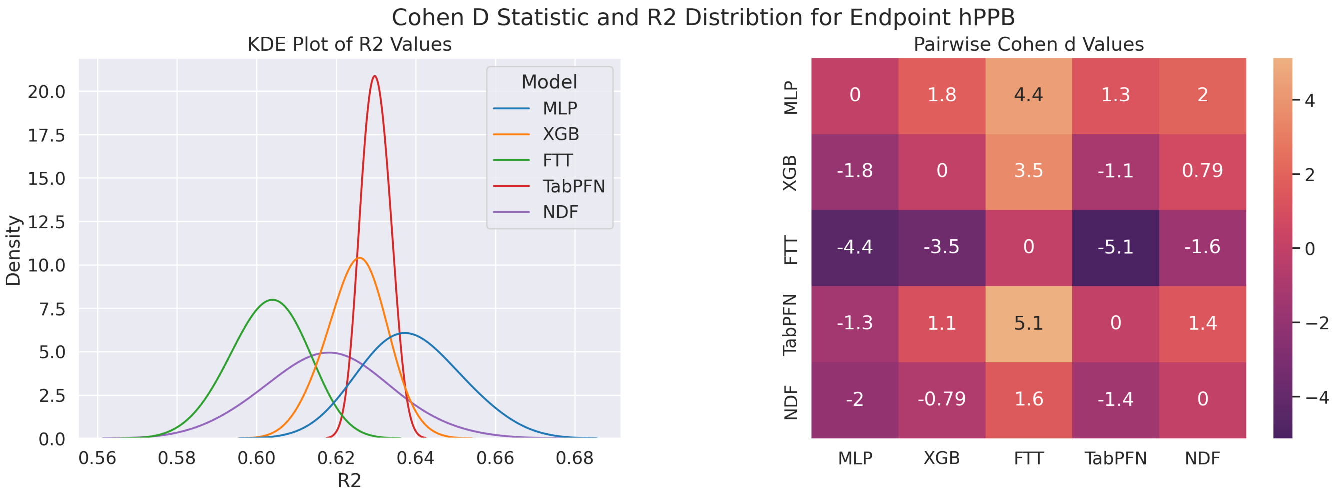
2.2.5 NDF

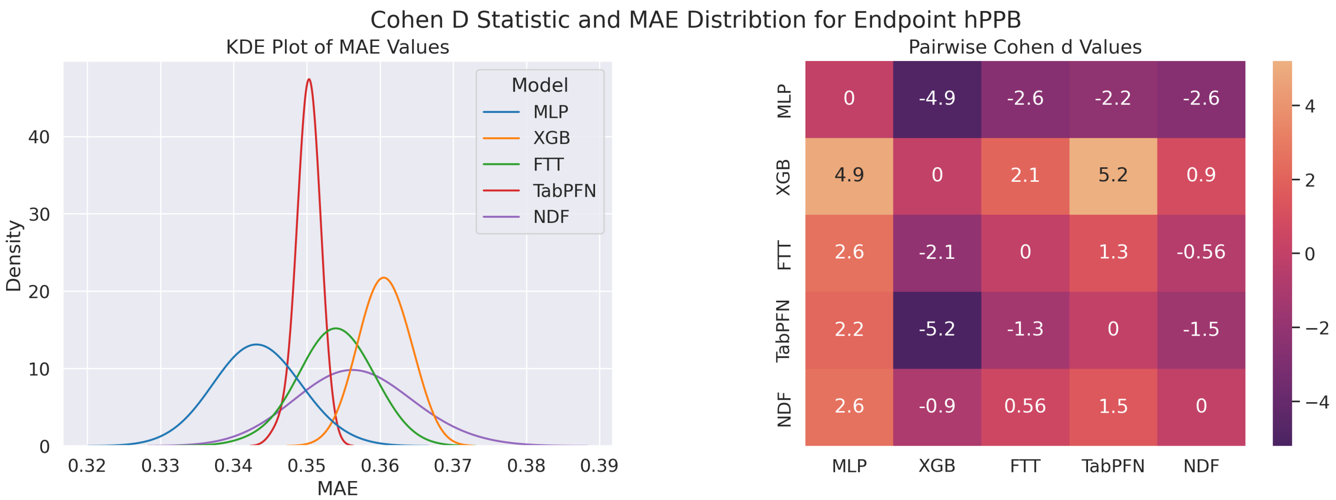
	0	1	2	3	4	5
params_batch_size	256	128	128	256	256	256
params_feature_rate	0.650000	0.550000	0.800000	0.900000	0.700000	0.900000
params_lr	0.000491	0.000515	0.000391	0.000780	0.000651	0.000413
params_n_tree	62	48	56	95	75	57
params_tree_depth	2	1	2	2	2	2
R2	0.514620	0.455917	0.327638	0.436656	0.402327	0.532735
Endpoint	hPPB	rPPB	Sol	RLM	HLM	MDR1_ER

3 Results

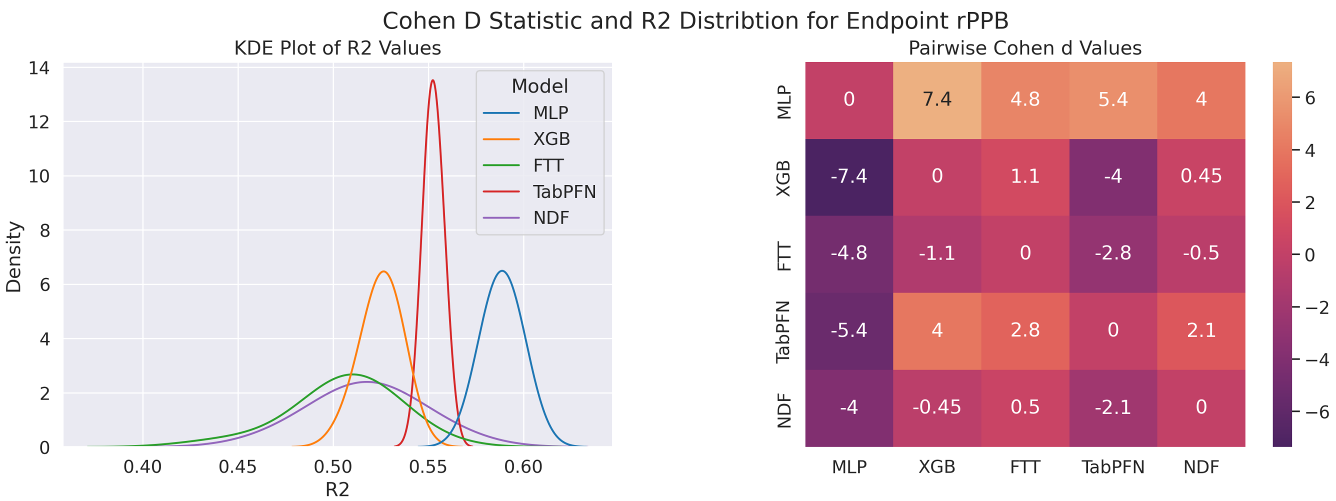
3.1 Test Performance per Endpoint

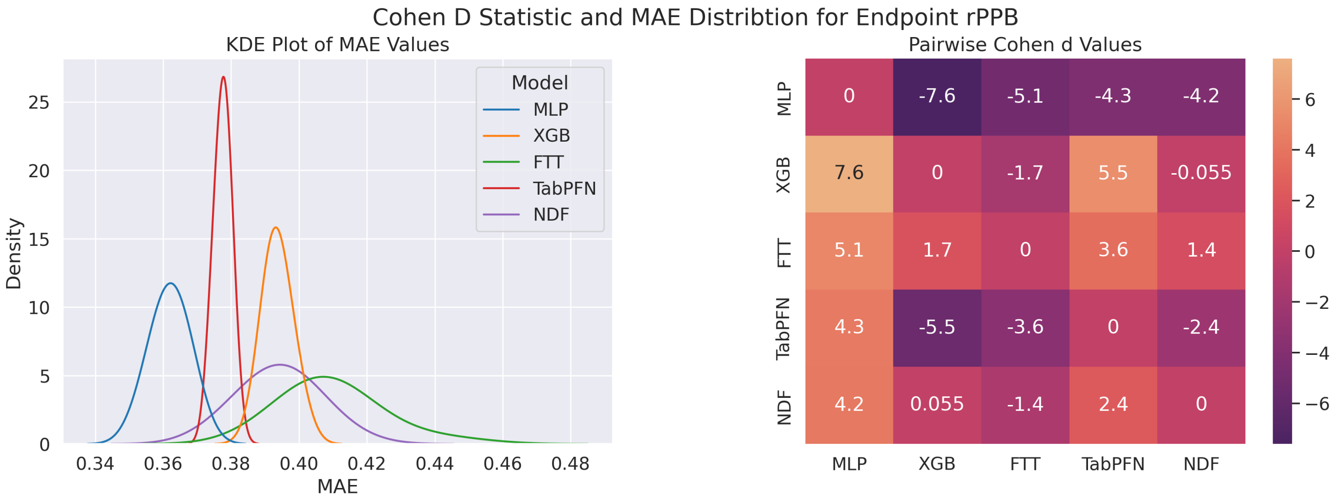
3.1.1 hPPB



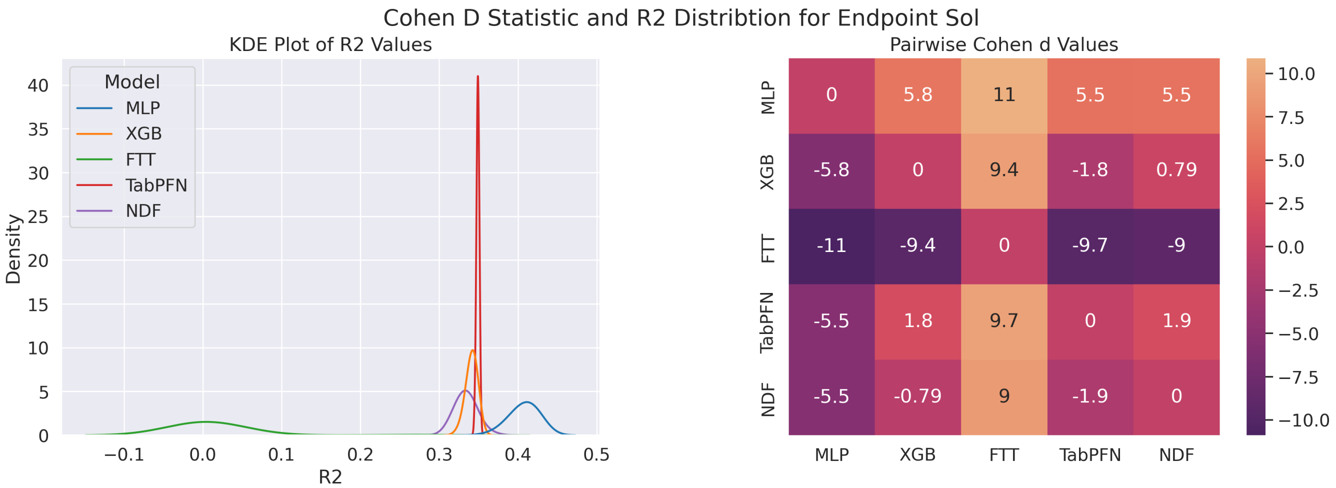


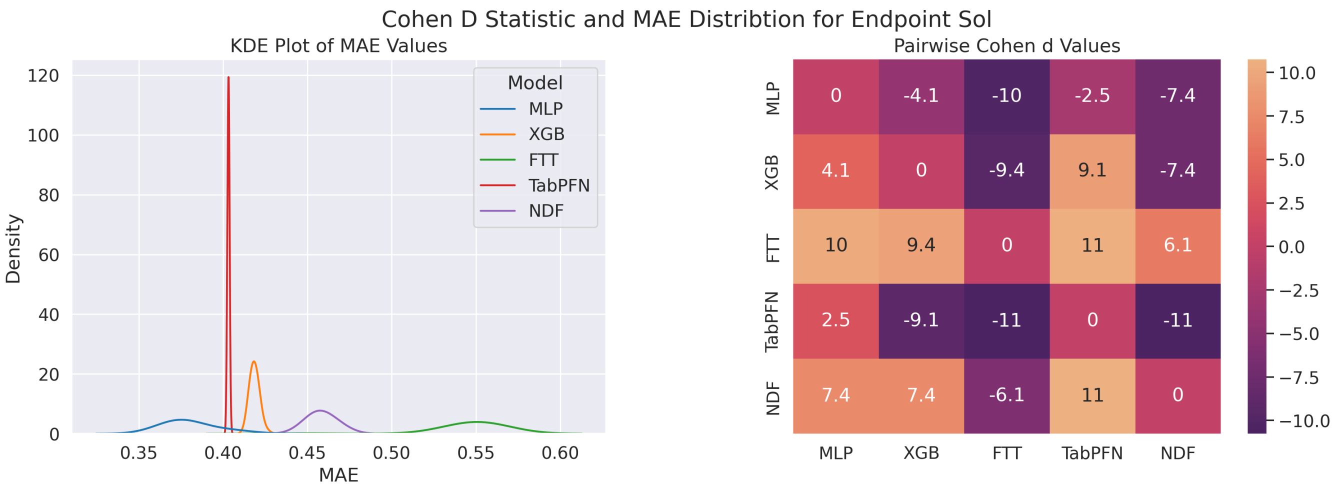
3.1.2 rPPB



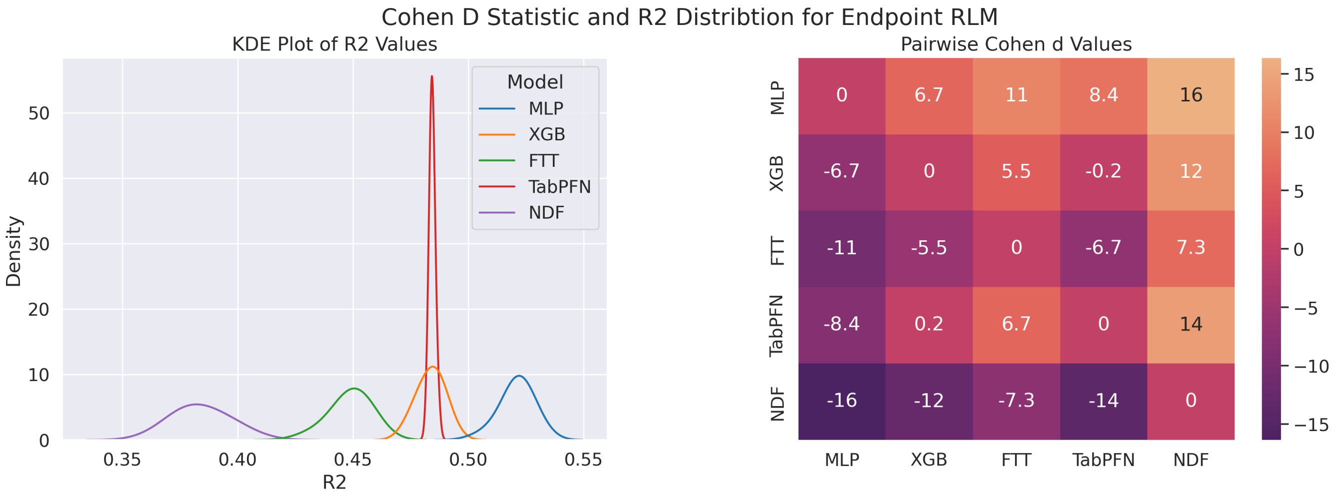


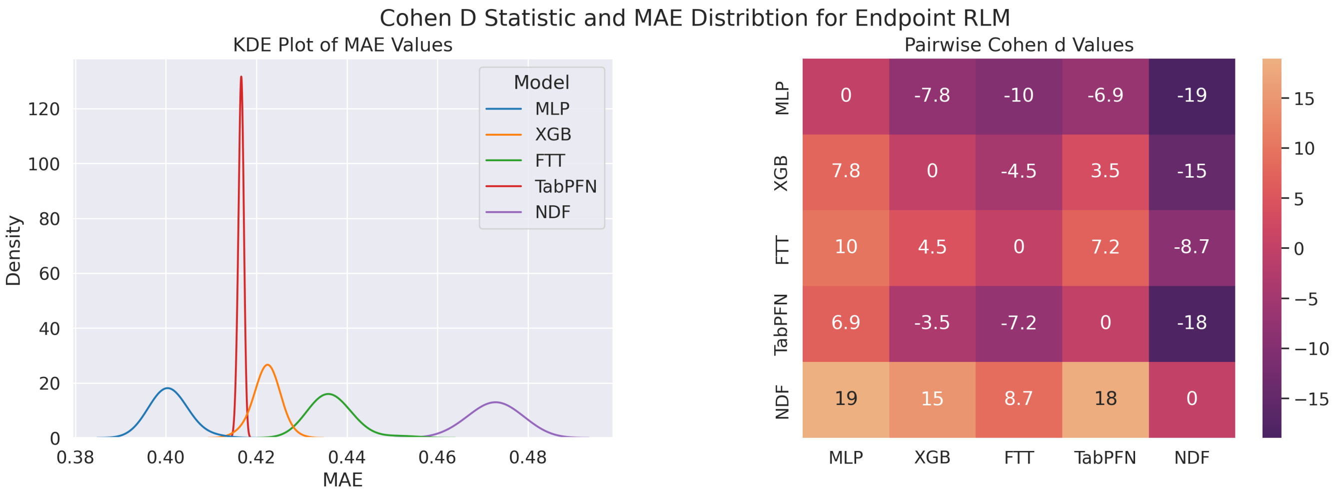
3.1.3 Solubility



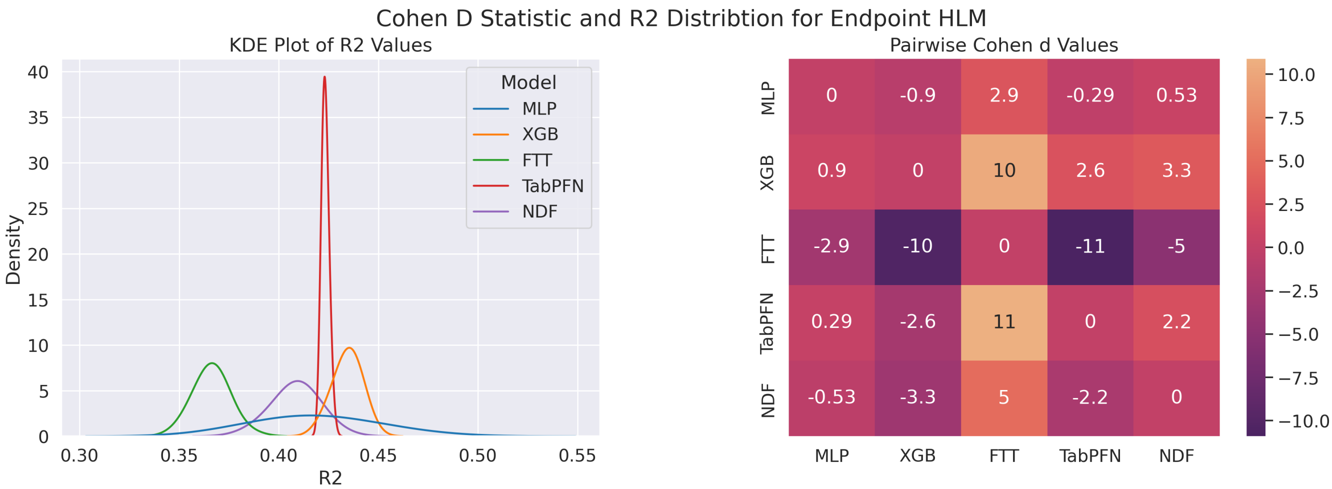


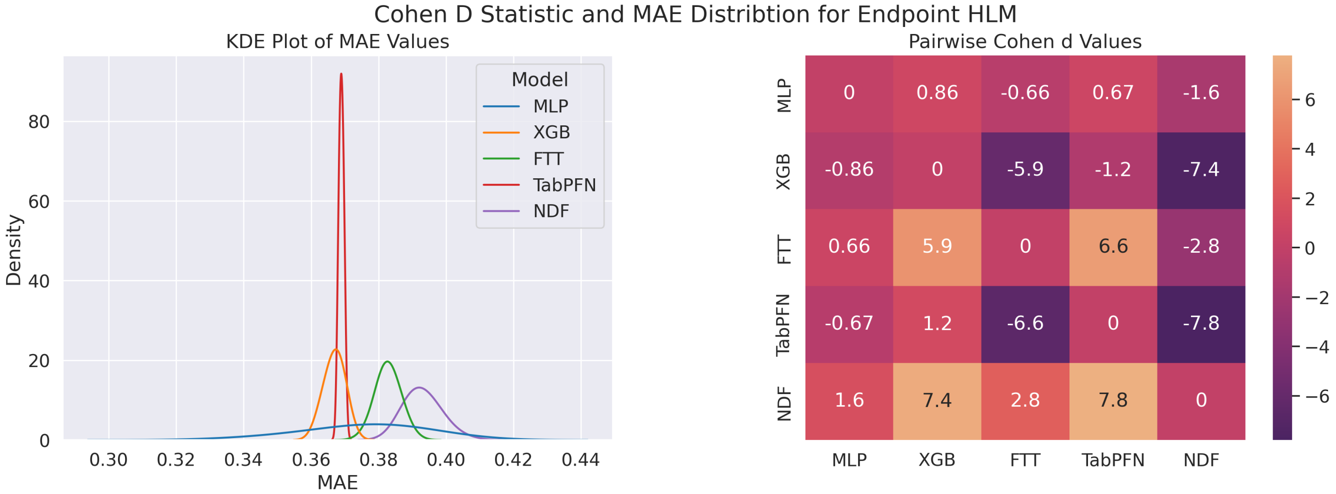
3.1.4 RLM



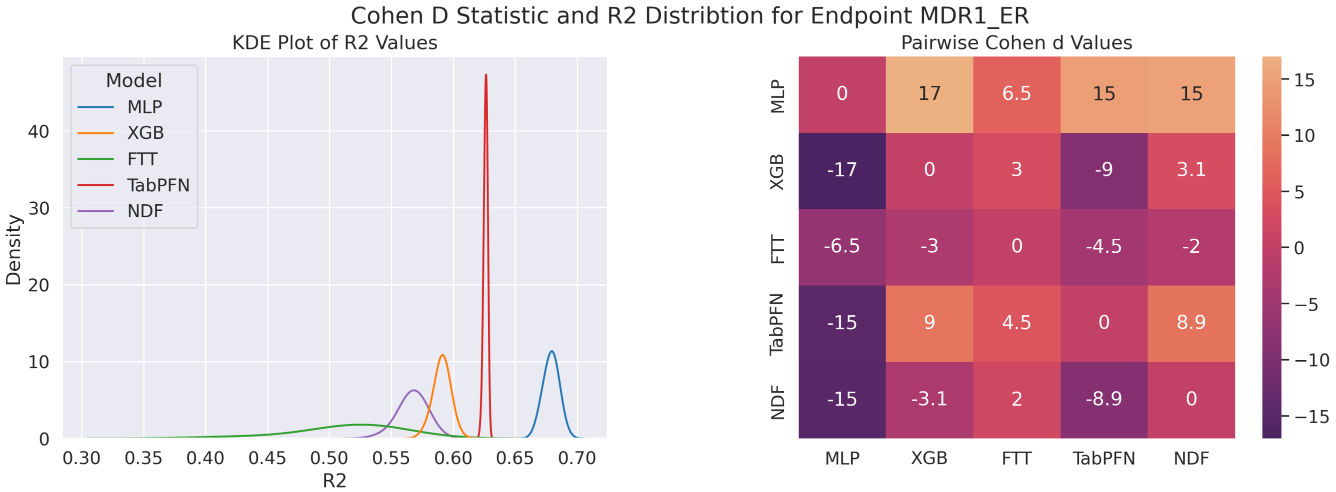


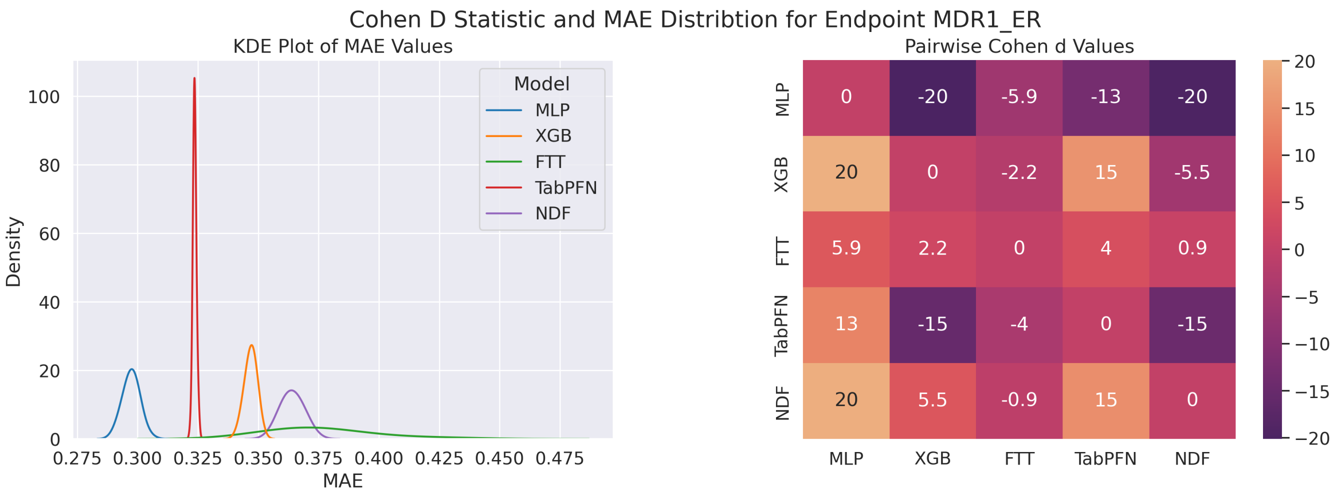
3.1.5 HLM




Cohen D Statistic and MAE Distribtion for Endpoint HLM
KDE Plot of MAE Values
Pairwise Cohen d Values
Model
MLP
XGB
FTT
TabPFN
NDF
Density
MAE
0 0.86 -0.66 0.67 -1.6
-0.86 0 -5.9 -1.2 -7.4
0.66 5.9 0 6.6 -2.8
-0.67 1.2 -6.6 0 -7.8
1.6 7.4 2.8 7.8 0

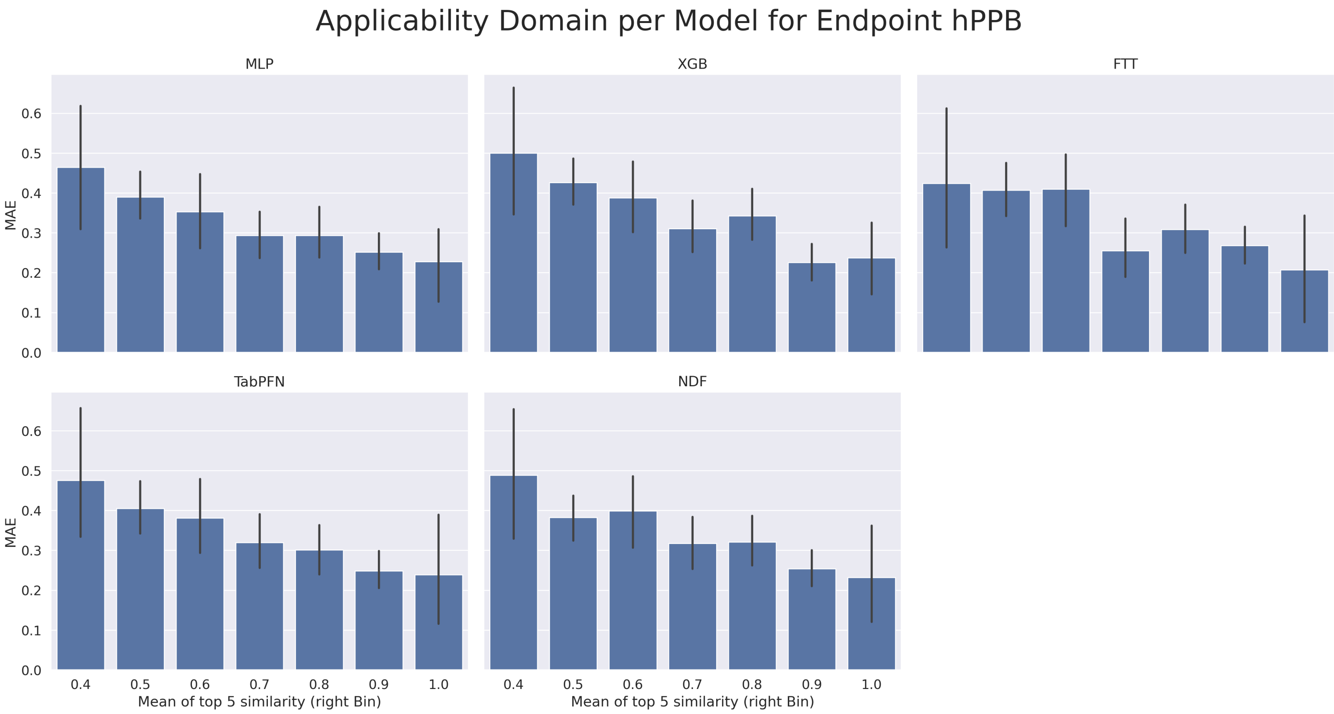
3.1.6 MDR1-ER


Cohen D Statistic and R2 Distribtion for Endpoint MDR1_ER
KDE Plot of R2 Values
Pairwise Cohen d Values
Model
MLP
XGB
FTT
TabPFN
NDF
Density
R2
0 17 6.5 15 15
-17 0 3 -9 3.1
-6.5 -3 0 -4.5 -2
-15 9 4.5 0 8.9
-15 -3.1 2 -8.9 0

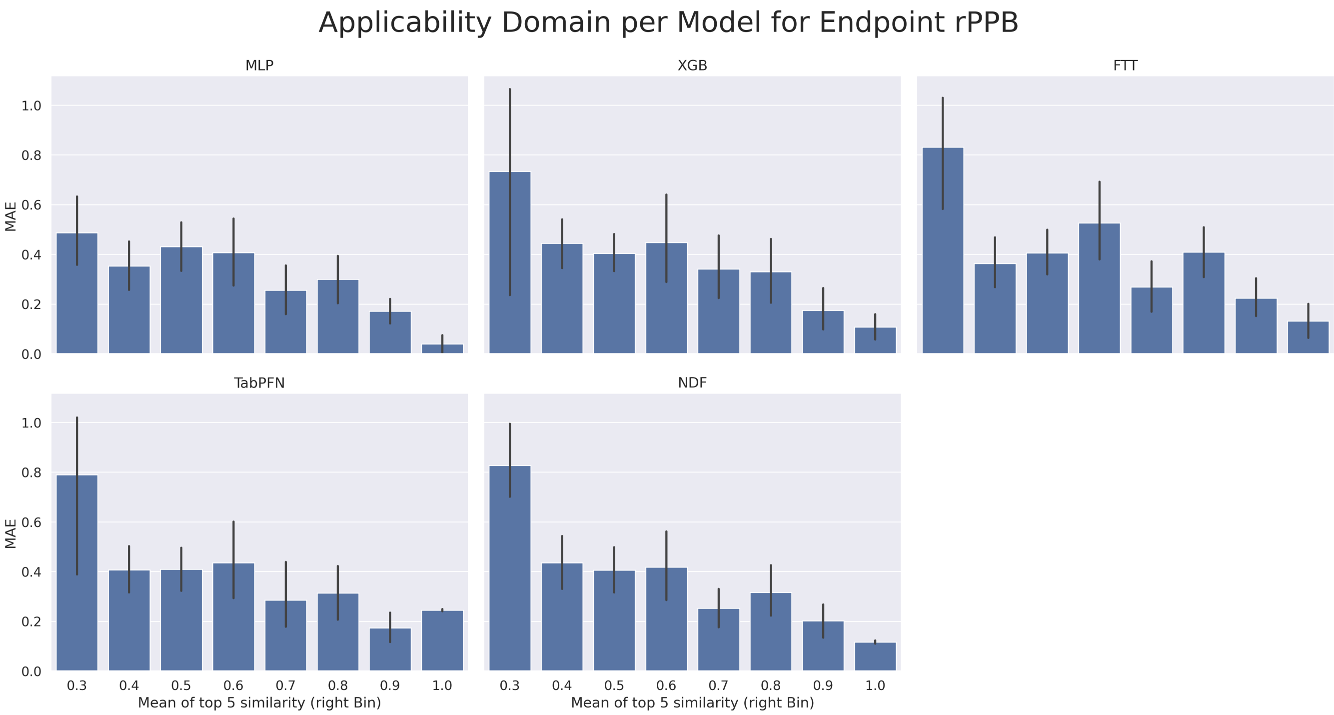

Cohen D Statistic and MAE Distribtion for Endpoint MDR1_ER
KDE Plot of MAE Values
Pairwise Cohen d Values
Model
MLP
XGB
FTT
TabPFN
NDF
Density
MAE
0 -20 -5.9 -13 -20
20 0 -2.2 15 -5.5
5.9 2.2 0 4 0.9
13 -15 -4 0 -15
20 5.5 -0.9 15 0

3.2 Applicability Domain

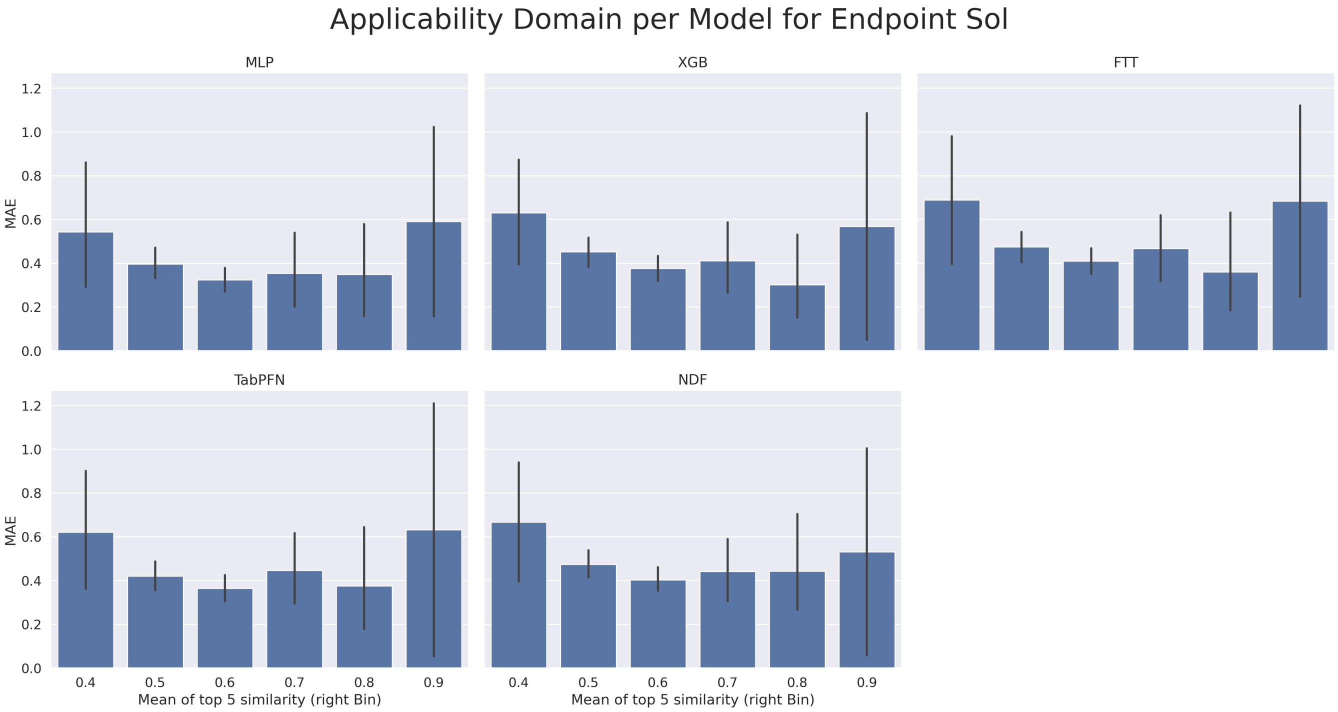
3.2.1 hPPB



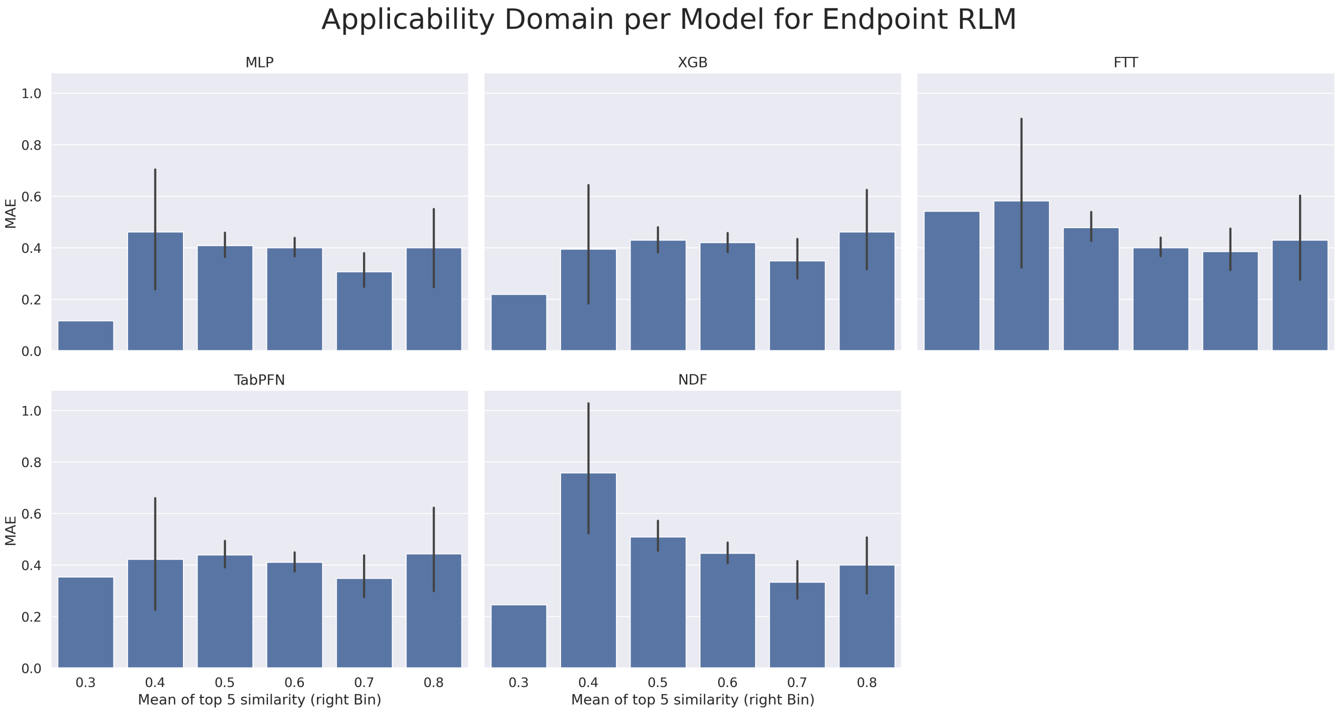
3.2.2 rPPB



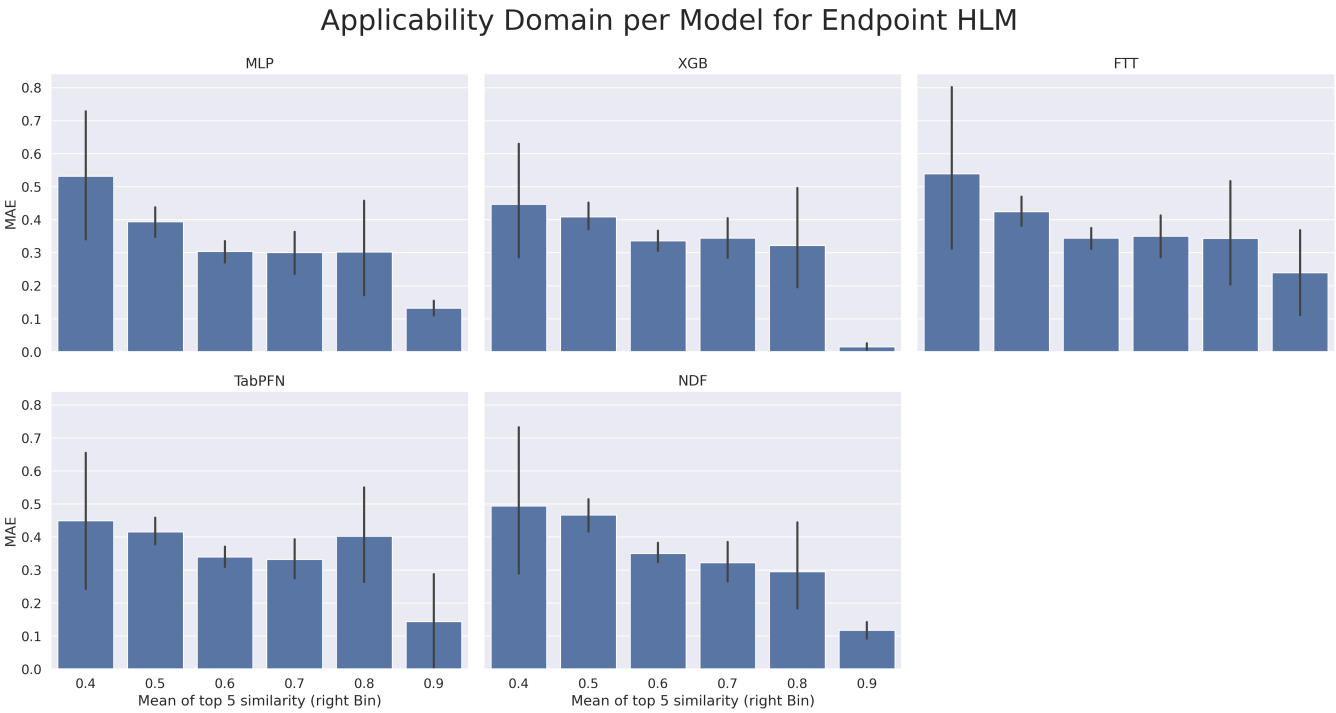
3.2.3 Solubility



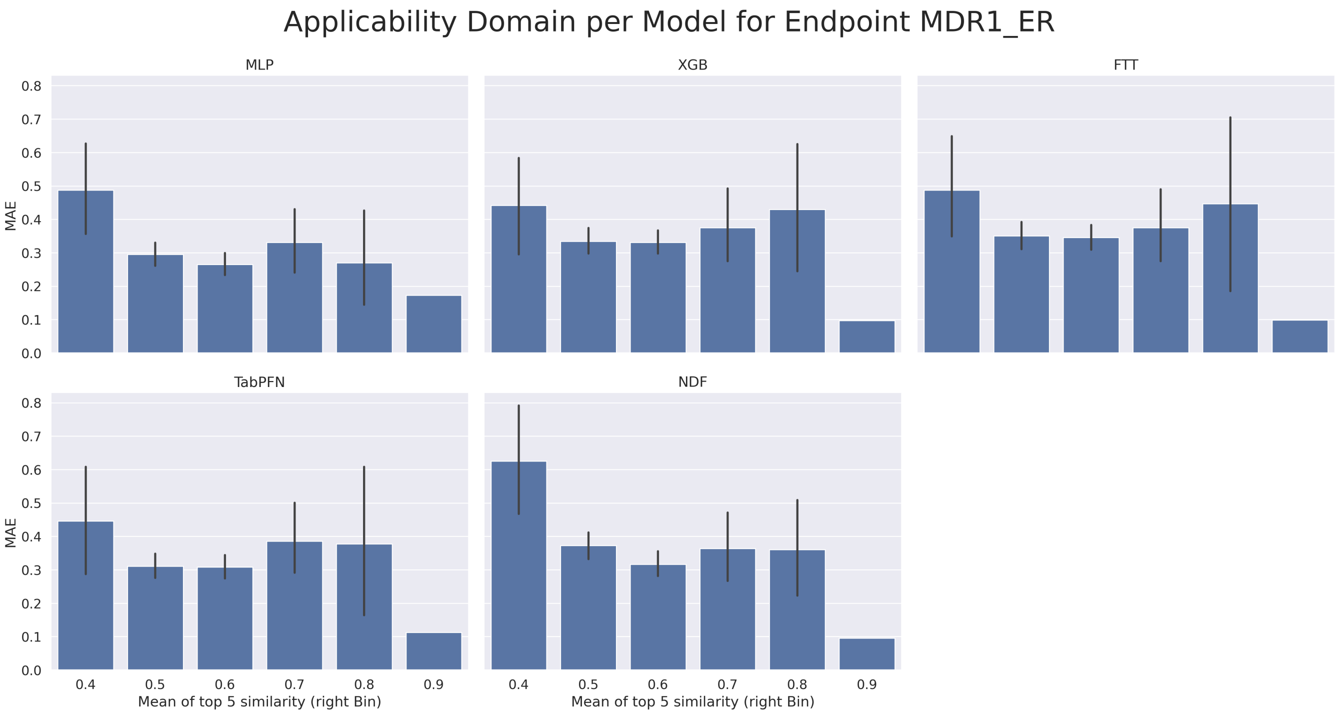
3.2.4 RLM



3.2.5 HLM



3.2.6 MDR1-ER



3.3 Dataset size and Performance

	Model	Endpoint	R2	MAE	count
0	NDF	HLM	0.408444	0.393044	3087
1	NDF	MDR1_ER	0.567167	0.364007	2642
2	NDF	RLM	0.384116	0.472673	3054
3	NDF	Sol	0.333980	0.458386	2173
4	NDF	hPPB	0.617922	0.356570	1808
5	NDF	rPPB	0.517290	0.394077	885
6	MLP	HLM	0.418340	0.375901	3087
7	MLP	MDR1_ER	0.678981	0.297322	2642
8	MLP	RLM	0.521011	0.400738	3054
9	MLP	Sol	0.407580	0.380065	2173
10	MLP	hPPB	0.639024	0.343249	1808
11	MLP	rPPB	0.588841	0.361912	885
12	FTT	HLM	0.366196	0.382793	3087
13	FTT	MDR1_ER	0.514749	0.376230	2642
14	FTT	RLM	0.448810	0.436665	3054
15	FTT	Sol	0.013637	0.546950	2173
16	FTT	hPPB	0.602909	0.353853	1808
17	FTT	rPPB	0.505550	0.409710	885
18	TabPFN	HLM	0.423355	0.368960	3087
19	TabPFN	MDR1_ER	0.626152	0.323651	2642
20	TabPFN	RLM	0.484334	0.416583	3054
21	TabPFN	Sol	0.349094	0.403110	2173
22	TabPFN	hPPB	0.629923	0.350155	1808
23	TabPFN	rPPB	0.552698	0.377532	885
24	XGB	HLM	0.434384	0.366888	3087
25	XGB	MDR1_ER	0.590831	0.346865	2642
26	XGB	RLM	0.483635	0.422276	3054
27	XGB	Sol	0.341131	0.418536	2173
28	XGB	hPPB	0.625117	0.360606	1808
29	XGB	rPPB	0.525305	0.393678	885

References

[1] L. McInnes, J. Healy, and J. Melville, "Umap: Uniform manifold approximation and projection for dimension reduction," *arXiv preprint arXiv:1802.03426*, 2018.

[2] L. Van der Maaten and G. Hinton, "Visualizing data using t-sne.," *Journal of machine learning research*, vol. 9, no. 11, 2008.

[3] S. Wold, K. Esbensen, and P. Geladi, "Principal component analysis," *Chemometrics and intelligent laboratory systems*, vol. 2, no. 1-3, pp. 37–52, 1987.